Statistics-Conditioned AI/ML CSI Feedback for Practical 5G MU-MIMO Transmission

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Abstract

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Accurate channel state information (CSI) is essential for practical multi-user multiple-input multiple-output (MU-MIMO) transmission in 5G New Radio (NR), since CSI quality directly affects user grouping, precoding, link adaptation, and overall cell throughput. Although Type 2 codebook-based CSI feedback provides a more expressive practical representation than Type 1, it still suffers from a significant feedback-overhead versus fidelity tradeoff, especially in interference-sensitive MU-MIMO operation. To address this limitation, this thesis proposes a statistics-conditioned AI/ML CSI feedback framework and evaluates it within an end-to-end radio access network (RAN) digital twin.

The suggested framework is based on an OAI-Sionna-based RAN twin in the context of single-cell 4T4R, with OpenAirInterface representing the practical 5G protocol stack and Sionna providing environment-aware channel generation. Three representative propagation conditions (UMi-LOS, UMa-LOS, and UMa-NLOS) are investigated with the help of a statistical system-level simulation model based on 3GPP TR 38.901. On this platform, Type 2 CSI processing is implemented to be practical, semi-orthogonal user selection with proportional-fair scheduler and regularized zero-forcing precoding are also used to facilitate end-to-end MU-MIMO transmission.

Not only is this practical transmission structure built up, but also a fundamental CsiNet-based encoder and an introduced conditioned CsiNet-based encoder are created to compress the CSI. The proposed solution, in contrast to the blind base encoder, utilizes long-term channel descriptions (based on the digital twin) as auxiliary information with respect to the encoding of CSI. Experimental results demonstrate that the suggested encoder reaches lower normalized mean square error on several compression ratios and propagation situations. End-to-end evaluation goes on to prove that the proposed method outperforms standard Type 1 SU-MIMO, Type 2 SU-MIMO, Type 2 MU-MIMO and blind CsiNet-based MU-MIMO in terms of block error rate, average MCS index and cell downlink throughput.

In the analyzed scenarios, the greatest improvement can be noticed in the UMa-NLOS setting, which has a more scattered channel structure and blind instantaneous-channel compression is not as effective. Such findings show that long-term statistical conditioning enhances the practical applicability of CSI when the feedback is constrained and gives significant system-level benefits in practical conditions of MU-MIMO transmission.

Overall, this thesis shows that CSI feedback can be improved significantly when the compression problem is formulated not as blind instantaneous-channel encoding, but as statistics-conditioned channel representation. By incorporating long-term channel information into a practical AI/ML CSI feedback framework, the proposed method improves the usefulness of CSI under limited

feedback and yields meaningful system-level gains in realistic MU-MIMO transmission. These results suggest that statistics-conditioned CSI feedback is a promising direction for practical 5G and beyond-5G MIMO systems.

Keywords : **5G NR, MU-MIMO, Type 2 PMI, Precoding, Scheduling, AI/ML CSI Compression**

Chapter 1.

Introduction

Massive multiple-input multiple-output (MIMO) is among the most significant enabling technologies of 5G New Radio (NR) due to its ability to enhance the spectral efficiency by means of spatial multiplexing and beamforming. In a real-world cellular network, a gNB with several antennas may direct the energy at desired users and minimize leakage to other users. This gain is manifested in the form of increased throughput of a single UE in SU-MIMO mode, and when using the same spatial degrees of freedom to serve multiple UEs simultaneously on the same time-frequency resource, it serves to increase overall cell throughput without the need to add more spectrum.

The channel state information (CSI) required to operate effectively in MIMO systems is extremely dependent on the precision of the channel state information (CSI) that is accessible in the gNB. During FDD operation, the downlink channel cannot be estimated using uplink reciprocity and thus the UE has to estimate the downlink channel using CSI reference signals and report it to the gNB in a compressed form. The 3GPP NR framework hence standardizes the procedure of CSI reporting by RI, PMI, and CQI along with various types of codebook structures, including the Type 1 and Type 2. Type 1 has relatively low feedback overhead and limited channel representation capability, but Type 2 has richer channel representation at the expense of much larger feedback overhead [2-4].

This tradeoff is particularly significant in MU-MIMO. In SU-MIMO, the main impact of the CSI error is on the beamforming gain of rank-adaptation performance of one single user. With MU-MIMO, however, the CSI error can have a more severe and systematic effect as it has direct implications on the suppression of inter-user interference. Users with channel directions that are sufficiently compatible to be transmitted simultaneously must be identified by the scheduler and leakage between selected users must be suppressed by the precoder. Poor user combinations may be chosen by the scheduler when the feedback CSI is wrong and the precoder might be unable to sufficiently null or regularize interference. Thus, in MU-MIMO, CSI quantization error is not merely decreasing signal gain, but turns into residual inter-user interference [5-8].

The practical MU-MIMO baseline of this thesis is based on the Semi-orthogonal User Selection (SUS) and Proportional-Fair (PF) scheduling and Regularized Zero-Forcing (RZF) precoding. SUS tries to select users who have channel directions well-separated, PF optimizes throughput and fairness, and RZF minimizes the interference and is more robust than the pure zero forcing when the CSI is not accurate. In this transmission chain, CSI errors influence several steps at once; they

corrupt the perceived geometry of user channels, alter their SUS choices, distort their estimation of effective rates to PF schedule, and impair the multi-user channel matrices used by the RZF precoder [5-8].

The classic method of codebook-based feedback cannot completely solve this problem. Feedback of type 1 is efficient with regard to overheads but tends to be too coarse to work effectively with interference sensitive MU-MIMO, particularly at high multipath richness or in multi-cluster channels. Feedback of type 2 is more expressive and more suitable to MU-MIMO, but it is still a structured and quantized approximation of the real channel. It must then be noted that in practice even the most effective traditional feedback mode has a non-negligible performance gap compared to ideal CSI, especially in MU-MIMO [2, 9, 10].

In such conditions, AI/ML-oriented CSI feedback is becoming an attractive avenue. Since CsiNet, many works have demonstrated that deep-learning-driven autoencoders are more effective at compressing CSIs compared to traditional baselines at the same level of compression rates. The approaches usually transform the channel to the angle-delay domain, whereby it is more structured, and learn a low dimensional latent representation in feedback [11-14].

Nevertheless, most of the existing AI/ML CSI-feedback schemes use a blind compression formulation: the encoder takes in just the instantaneous channel realization and has to deduce all possible relevant structure using this sample only. It is inefficient to do it in realistic wireless scenarios since the channel is not created with a completely unknown distribution at each reporting time. It is rather influenced by slow environmental structures like buildings, street plan, topography and position of users. Hence, much of the channel structure is temporally constant and can be used as side information [11-16].

The present observation suggests the introduction of statistics-conditioned CSI compression. Two types of long-term channel information are especially pertinent to this thesis: the spatial covariance matrix and the power delay profile (PDP). The covariance matrix represents the principal spatial eigenspace and the array-domain correlation whereas the PDP indicates the mean delay-domain energy distribution and it is an indicator of multipath richness and frequency selectivity. Both have much lower variances than instantaneous CSI, and so may be used as side information to decrease the uncertainty of the instantaneous channel and thus enhance the compression rate at a fixed feedback cost [15, 16].

A practical way to obtain such long-term statistics is through a Radio Access Network (RAN) digital twin. Recent work on wireless digital twins has shown that environment-aware virtual network platforms can support both execution-level evaluation and long-term radio knowledge extraction. In this thesis, a digital twin is not used merely as a simulator, but as a dual-purpose platform: it executes the actual end-to-end 5G protocol stack and simultaneously provides the long-term channel statistics later used by the proposed encoder [17–20].

To realize this idea, the thesis adopts an OAI-based RAN digital twin integrated with Nvidia Sionna's Ray Tracing(RT) channel model. OpenAirInterface(OAI) provides a realistic and modifiable implementation of the 5G RRC, MAC, and PHY stack, while Sionna RT provides GPU-accelerated experimental environment and ray-tracing-based environment-aware channel generation. This combination allows the thesis to generate MU-MIMO channels from actual urban geometry and to pass them through a practical protocol stack in which CSI measurement, feedback, scheduling, and precoding are all explicitly modeled [17–20].

The use of ray-tracing-based channels is especially important. Many prior CSI-feedback studies use stochastic channel models or abstract channel datasets, which are useful for compression benchmarking but do not always preserve the physically meaningful multi-user geometry that governs MU-MIMO behavior. By contrast, ray tracing produces channels shaped by buildings, streets, blockage, and user positions, and therefore naturally captures user-to-user spatial relationships. This is essential for evaluating the practical SUS+PF + RZF chain, because that chain depends directly on user-channel compatibility [1, 18, 19].

The thesis deals with three typical cases, i.e., UMa-NLOS, UMi-LOS, and UMa-LOS. These cases are physically different propagation situations. In UMa-NLOS, it frequently produces a larger angular support and richer reflected structure that could enhance user separability, but also adds to channel complexity. UM-LOS can often produce very strong directional components, which is easier to align a single-user beam to, but becomes more prone to directional CSI errors. UMa-LOS provides a propagation condition with a predominant LOS part as well as large-scale spatial variability associated with macro-cell deployment, and it can be used to test whether statistical conditioning would remain effective in large-scale urban environment with strong directional structure [1, 18, 19].

The gap between offline reconstruction quality and end-to-end wireless-system performance is another important issue. Numerous CSI compression studies have reported improvements in normalized mean square error (NMSE) or cosine similarity, however, these measures are not sufficient to estimate the ultimate effect on the system. In our operational MU-MIMO chain, there are multiple nonlinear processes between CSI reconstruction and throughput: SUS operates with geometric thresholds, PF scheduling relies on rate estimates, RZF has conditioning of the reconstructed multi-user channel matrix, and CQI/MCS determinations are quantized. Consequently, a small reconstruction benefit can provide a significant throughput benefit at one operation area and a small benefit in another. Which is why end-to-end evaluation is essential [17-20].

The research problem addressed in this thesis can therefore be summarized as follows. First, practical MU-MIMO in 5G NR is strongly limited by CSI quality. Second, Type 2 feedback improves upon Type 1 but still leaves a significant CSI-induced MU-MIMO gap. Third, existing AI/ML CSI encoders generally ignore the long-term environment-dependent structure of realistic channels. Fourth, a RAN digital twin provides a way to estimate those statistics and evaluate their practical value in a realistic protocol stack. The central hypothesis of this thesis is that if long-term

channel statistics extracted from the digital twin are used as conditioning information in an AI/ML CSI encoder, then the system can achieve more accurate CSI under the same effective feedback cost, and that this gain will translate into measurable MU-MIMO throughput improvement.

The first objective of this thesis is to build an OAI-based RAN digital twin that supports both realistic end-to-end execution and channel-statistics extraction. This includes the integration of OAI with a Sionna-based ray-tracing channel engine and the design of a centralized Core Emulator for reproducible experiment configuration and orchestration [17–20]. The second objective is to establish practical baselines corresponding to Type 1 SU-MIMO, Type 2 SU-MIMO, and Type 2 MU-MIMO. In particular, the MU-MIMO baseline is implemented using SUS+PF scheduling and RZF precoding so that CSI quality affects user pairing and interference suppression in a realistic and interpretable manner [5–8]. The third objective is to develop a statistics-conditioned AI/ML CSI encoder that extends a CsiNet-style baseline by explicitly incorporating covariance and/or PDP information extracted from the digital twin. Unlike blind neural compression, the proposed framework uses long-term channel structure as side information in both encoding and decoding [11–16]. The final objective is to show that the gain of the proposed encoder is not limited to offline reconstruction metrics, but appears as a measurable end-to-end performance gain in the realistic MU-MIMO chain implemented in the digital twin. This includes improvements in user pairing, precoding effectiveness, cell throughput, and spectral efficiency under the UMa-NLOS, UMi-LOS, and UMa-LOS scenarios [17–20].

The thesis will be centered on downlink MIMO operation of 5G NR FR1 Band 78 at 3.5GHz with the subcarrier spacing of 30kHz and 106 PRB. The realistic antenna range is 2T2R to 4T4R with dual-polarized mode, and the major evaluation goal is 4T4R since it can support practical Type 2 and MU-MIMO measurements and remains manageable in the existing combined RAN-twin deployment [1-4]. The propagation environments are a real-life urban environment produced by Sionna ray tracing, more precisely, UMa-NLOS, UMi-LOS, and UMa-LOS. SUS+PF is the scheduler and RZF is used as the downlink precoder. Such design decisions are deliberate since they specify a viable MU-MIMO chain which can be investigated in terms of its sensitivity to CSI quality and which is still interpretable in end-to-end experiments [5-8, 17-20].

The thesis has been arranged in this manner. Chapter 2 will discuss the MIMO channel model, the 5G NR CSI acquisition framework, the operational MU-MIMO baseline, and the other literature concerning codebook-based. Chapter 3 describes the architecture and implementation of the OAI-based RAN digital twin, including the Core Emulator, the Sionna channel proxy, and the MU-MIMO scheduler/precoder integration. Chapter 4 presents the CsiNet baseline and the proposed statistics-conditioned AI/ML CSI encoder together with the dataset design and offline evaluation methodology. Chapter 5 evaluates the proposed method in the end-to-end digital twin and compares it with the three practical baselines and blind CsiNet. Chapter 6 concludes the thesis.

Chapter 2.

MIMO Channel Model and 5G NR CSI Acquisition–Scheduling System

This chapter provides the technical background required for the remainder of the thesis and establishes the practical transmission framework used throughout this work. Its purpose is not merely to review standard MIMO concepts, but to explain, in a logically consistent manner, why CSI representation becomes a critical bottleneck in practical MU-MIMO operation and why an improved CSI feedback mechanism is needed. To this end, the chapter first introduces the downlink MIMO channel model and the realistic propagation environments considered in this work, with emphasis on the distinction between instantaneous channel realizations and long-term channel statistics. It then presents the scheduling system model adopted in this thesis, including downlink user selection, fairness-aware scheduling, and practical precoding, in order to clarify how CSI quality affects both user grouping and interference suppression in MU-MIMO transmission. After that, the chapter describes the conventional CSI feedback framework in 5G NR, focusing on CSI-RS-based measurement and Type 1 and Type 2 codebook-based reporting. Through these discussions, the chapter establishes the practical system context in which the proposed statistics-conditioned CSI encoder is later developed and evaluated.

2.1 MIMO Channel Model

2.1.1 MIMO System Model

We consider a downlink MIMO-OFDM system in which a gNB equipped with N_t transmit antennas serves one or more UEs, each with N_r receive antennas. For the k-th subcarrier, the received signal can be written as

$$y[k] = H[k]W[k]x[k] + n[k], \tag{2.1}$$

where $H[k] \in \mathbb{C}^{N_r \times N_t}$ is the channel matrix, $W[k]$ is the precoding matrix, $x[k]$ is the transmitted symbol vector, and $n[k]$ is complex additive noise. The given expression can be interpreted as the channel matrix defines the coupling between the gNB antenna array and the UE antenna array, and the precoder decides how the transmitted streams map into the spatial domain. Because the role of CSI feedback is to allow the gNB to select an adequate $W[k]$, the accuracy in how the UE reports its channel directly affects the overall MIMO performance.

The precoder in a single-user MIMO system is optimized to serve only one UE at any given time, and the goal is to achieve the highest possible data rate or transmission reliability of the user. Multi-user MIMO systems, on the other hand, schedule several UEs with different channels at the same time and frequency resource, and then the precoder should be able to control all the channels collectively. Hence, the accuracy specification of CSI is significantly tighter in MU-MIMO compared to SU-MIMO. The error in a channel estimate in SU-MIMO mostly leads to a decrease in beamforming gain or inappropriate rank adaptation. An imperfect channel estimate in MU-MIMO, however, alters the inferred relationship between users and may result in wrong pairing of users and poor interference reduction. This separation is what is fundamental to the thesis.

The channel has a multi-timescale structure. A channel realization H is fast changing due to small scale fading, phase rotation, and mobility and may change at the slot level or even symbol level in case of high user speeds. On the other hand, the long-term channel statistics have a much-lower rate of variation and are determined by the propagation environment, the antenna geometry and the position of the user. One common description of this long-term structure is the covariance matrix

$$R_H = \mathbb{E}[vec(H)vec(H)^H], \tag{2.2}$$

which captures the dominant spatial eigenspace and the correlation structure of the channel. Another important descriptor is the power delay profile (PDP), which characterizes the average power distribution over delay taps and therefore reflects the multipath richness and delay-domain sparsity of the propagation environment.

These statistics are important in the present thesis because they contain side information about how the channel is "typically shaped" in a given environment. If such information is known, the uncertainty of the instantaneous channel is reduced, and the channel can be compressed more efficiently. This observation provides the theoretical basis for the statistics-conditioned CSI encoder proposed later in the thesis. In other words, the proposed method is based on the idea that the encoder should not treat every channel realization as if it were generated from an unknown distribution; instead, it should exploit prior knowledge of the channel statistics associated with the environment.

2.1.2 3GPP Channel Model and Sionna

The 3GPP TR 38.901 channel model defines representative deployment scenarios such as Urban Macro (UMa), Urban Micro (UMi), and Rural Macro (RMa), together with associated propagation parameters including path loss, shadow fading, delay spread, angular spread, and line-of-sight probability. These models have been widely used to evaluate wireless communication techniques because they offer a standardized compromise between realism and tractability. However, for the purpose of this thesis, the role of the channel model is not limited to providing average path-

loss or delay-spread parameters. The channel model must preserve the spatial geometry of multi-user propagation so that CSI feedback, scheduling, and MU-MIMO interference can be meaningfully studied.

To achieve this, the present thesis uses Sionna as the channel-generation engine. Nvidia Sionna provides both 3GPP-compliant stochastic channel models and ray-tracing-based propagation modeling in a GPU-accelerated framework. In this work, Sionna is used not only as a channel emulator for end-to-end system execution but also as a data-generation platform for AI/ML training. This is an important point: the same channel-generation engine is used both to create the channel conditions under which the protocol stack operates and to extract the long-term statistics later used as conditioning information for the proposed encoder.

The channel in that thesis is based on realistic environment-conscious propagation. The channel is created by urban geometry such as buildings and landscape with the help of Sionna-ray-tracing to calculate physically meaningful multipath components. Therefore, the channel matrices obtained in this work are not independent random matrices; they are environment-conditioned realization. The importance of this characteristic is most significant in MU-MIMO, because the relative spatial positions between the users are determined by shared environmental architecture. Practical, two users may be situated in the same geometric sectors or may encounter the same reflection paths, and this would cause their channels to be highly correlated, making them hard to spatially multiplex.

Three representative situations are discussed in the thesis: UMa-NLOS, UMi-LOS, and UMa-LOS. The UMa-NLOS scenario usually has greater angular spread and more diverse reflected multipath owing to the use of macro-cells and lack of line-of-sight. These tend to increase complexity of the channel yet may enhance user separability since users become less controlled by a shared LOS beam. The UMi-LOS scenario, by contrast, typically has a very powerful dominant path and a more compact angular shape. Although this can simplify individual beam alignment, it can also lead to more similar user channels when users are in the same street corridor, which is bad for MU-MIMO matching. The UMa-LOS scenario is an urban macro-cell environment with a dominant line-of-sight component. Its propagation characteristics are generally between that of UMi-LOS and UMa-NLOS: it is more spatially variable than UMi-LOS, due to its larger cell size, and more directional than UMa-NLOS, as a result of LOS propagation. As a result, it provides a meaningful intermediate operating condition for evaluating CSI compression and MU-MIMO scheduling.

Another important characteristic of realistic wireless channels is sparsity in transformed domains, especially the angle-delay domain. Much of the channel energy is concentrated in a relatively small number of dominant components, which is one of the reasons codebook-based and AI-based compression methods are effective. However, the average distribution of this sparsity pattern is not arbitrary; it is shaped by the long-term environment. The covariance matrix and PDP can therefore be interpreted as descriptors of the average angle-delay energy structure. If this

structure is provided to the encoder in advance, the encoder can allocate its limited feedback capacity more efficiently to the part of the channel that truly varies from realization to realization.

2.2 MU-MIMO Scheduling System Model

In SU-MIMO, the scheduler assigns multiple layers to one UE in order to maximize that UE's throughput. The precoder is optimized only for the channel of the selected UE, and the main challenge is to find a beam or set of layers that matches the user's channel subspace. The effect of CSI error in SU-MIMO is therefore mainly a reduction in beamforming gain, stream separation quality, or link adaptation accuracy.

MU-MIMO has a completely different nature. In MU-MIMO, several UEs occupy the same time-frequency resource at the same time. The gNB needs to determine which users are to be co-scheduled and it needs to calculate a precoding matrix that will cancel out the interference between these users. To achieve this, precise information about every individual user channel as well as the relative geometry of the channels between the chosen users is necessary. When the direction of the channels is not precise enough, the scheduler can select poorly matched user pairs, and the precoder may not null or sufficiently reduce the noise level. Hence, the error of the CSI in the MU-MIMO turns into the residual inter-user interference which is usually significantly more destructive than the beamforming loss associated with the SU-MIMO.

The practical MU-MIMO baseline used in this thesis is the scheduling of SUS+PF and the precoding of RZF. SUS finds out if the users are spatially similar or not through the similarity of their channel directions, PF tries to balance throughput and fairness, and RZF constructs the final precoder based on the selected set of users. In this link, CSI error has dual-effect. It is corrupting user pairing as an illusion of orthogonality between users, and it is corrupting the effective channel matrix that enters the RZF stage. It is the reason why more accurate CSI at the same feedback budget is not just a link-level facility but a fundamental demand behind opening up MU-MIMO benefits.

Theoretically speaking, optimal CSI would make it possible to use MU-MIMO to realize significant increases in cell capacity over SU-MIMO since multiple users could be provided almost orthogonally with near-orthogonal beams. Practically speaking, finite and quantized feedback limits these benefits. Hence, the primary issue of concern in this thesis may be formulated in the following way: what is the method with which the system may communicate more precise CSI with the same or similar amount of bits, so that the realistic MU-MIMO link could come closer to its hypothetical operation?

2.2.1 Downlink Transmission and User Set Selection

The downlink transmission considered in this thesis is based on a practical multi-user scheduling framework in which a gNB selects a subset of active UEs for transmission in each scheduling interval. Let $\mathcal{U} = \{1, 2, \dots, U\}$ denote the set of active users in the cell. At each transmission interval, the scheduler chooses a subset $\mathcal{S} \subseteq \mathcal{U}$, where $|\mathcal{S}| = 1$ corresponds to SU-MIMO operation and $|\mathcal{S}| > 1$ corresponds to MU-MIMO operation.

For a selected user set $\mathcal{S}$, the transmitted signal on subcarrier k is written as

$$x_{tx}[k] = W_{\mathcal{S}}[k] s_{\mathcal{S}}[k], \tag{2.3}$$

where $W_{\mathcal{S}}[k]$ denotes the precoding matrix associated with the selected users and $\mathbf{s}_{\mathcal{S}}[k]$ denotes the stacked data symbol vector for those users. The received signal at user $u \in \mathcal{S}$ is then given by

$$y_u[k] = H_u[k] W_{\mathcal{S}}[k] s_{\mathcal{S}}[k] + n_u[k] \tag{2.4}$$

The expression clarifies that the downlink performance of user $\mathcal{U}$ is not only dependent on its own channel $\mathrm{H}_u[k]$, but also on the chosen user set $\mathcal{S}$ and the consequent multi-user precoder. Hence, the scheduling issue and the precoding issue are naturally connected.

The scheduler cannot consider all available user subsets and all possible precoding matrices in a real system. Rather, it has to use an acceptable decision rule that takes into account the existing CSI in deciding which users can be served efficiently with each other. On the system discussed in this thesis, this is done by a combination of fairness-sensitive utility evaluation and spatial compatibility evaluation. This is a practical implication that the scheduler should not only maximize instantaneous throughput but should also keep a fair balance at a reasonable level as well as utilize the spatial multiplexing benefit of MU-MIMO [5, 6, 31–33].

One of the most significant characteristics of the scheduling model applied in this work is that the choice is made on practical CSI rather than on perfect full channel knowledge. The efficient channel directions that are employed in scheduling are reorganized using the acquisition and feedback mechanism of the CSI discussed subsequently in this chapter. Therefore, the scheduler has the same information constraint as a real-life 5G NR system. It is significant since the main aim of the thesis is to explore the impact of better CSI representation on not just channel reconstruction accuracy but also on the final scheduling performance.

2.2.2 Utility Metric and Semi-Orthogonality Criterion

The scheduling model used in this thesis will incorporate proportional-fair utility and a semi-orthogonality condition. These two factors represent two different requirements of practical MU-MIMO scheduling. One is that it must use positive channel conditions without starving weaker users

too much. Two is that the users chosen to transmit simultaneously must be sufficiently distant in space so that linear precoding may reduce inter-user interference.

The proportional-fair mechanism allocates a utility to every user based on the ratio of its available rate and its historical average served rate. To use u, it could be written as

$$M_u = \frac{R_u^{inst}}{\bar{R}_u}, \tag{2.5}$$

It is represented by R_u^{inst} as the possible instantaneous rate estimate and $\bar{R}_u$ as the long-term average throughput of the respective user. Such a measure is beneficial to those users that are in a better state of affairs than they were in their last service experience. Consequently, it ensures some form of balance between opportunistic scheduling and equity across time [31–33].

Nevertheless, high utility is insufficient to achieve MU-MIMO transmission. Two users may have good PF parameters regardless of whether their channel directions are similar or not. This is why the scheduler also considers spatial compatibility between candidate users based on a semi-orthogonality metric.

We will use h_i and h_j to represent effective channel-direction vectors of the users i and j, respectively. respectively. The normalized similarity is expressed as

$$\rho_{ij} = \frac{\left|h_i^H \mathrm{h}_j\right|}{\|\mathrm{h}_I\|\|\mathrm{h}_j\|} \tag{2.6}$$

When ρ_{ij} is very low, the users can be approximated as being orthogonal and simultaneous transmission can be advantageous. When ρ_{ij} is high, the users are in comparable spatial directions and thus cannot be good MU-MIMO partners. This criterion is used by the scheduler to filter or to rank candidate user combinations [5, 6].

These two principles together give a pragmatic scheduling approach. Initially, the evaluation of the users is based on their proportional-fair utility. Secondly, the scheduler finds the pairs of high-utility users that meet the semi-orthogonality requirement. Thus, MU-MIMO opportunity can exist only with the fulfillment of both conditions: the users should be valuable to serve (throughput fairness) and spatially compatible.

This structure is particularly important in the context of this thesis because it makes CSI quality directly relevant to the scheduler. If the effective channel directions are inaccurately represented, then the scheduler may misjudge the value of ρ_{ij}. Consequently, it might choose a set of two users that would seem almost orthogonal in the practical CSI, yet are extremely interfering on the actual channel, or could also fail to accept a valid MU-MIMO opportunity, as the channel relation is not

preserved appropriately in the feedback representation. Therefore, the precoder is not the only factor that is affected by the quality of the CSI, but also the scheduling decision.

2.2.3 Regularized Zero-Forcing (RZF) Precoding

Once the user set is chosen, regularized zero-forcing is applied to compute the downlink precoder. The reason why RZF is adopted in this thesis is that it provides a reasonable trade-off between interference suppression and resistance to imperfect CSI. Zero forcing can be successful in cases where the channel is precisely known and the selected users have been strongly isolated, but it fails to be stable in the presence of an effective channel matrix with noise, distortion or poor conditioning. With the addition of a regularization term, RZF overcomes these drawbacks and is thus better suited to be used in a practical limited-feedback implementation [7, 34–37].

Let H_S represent effective multi-user channel matrix obtained by stacking the direction vectors of the chosen user set S. The RZF precoder is given by

$$W = \mathrm{H}_S{}^H\left(\mathrm{H}_S\mathrm{H}_S{}^H + \alpha I\right)^{-1}, \tag{2.7}$$

where α is the regularization factor. As α tends towards zero, the solution tends towards strict zero forcing; as α is increases, the solution has better resistance to numerical instability and CSI uncertainty. The matrix H_S in the practical system discussed in this thesis is not based on perfect CSI, but instead on the practical representation of CSI accessible to the gNB. Hence, the quality of CSI will directly affect the quality of the final precoder.

The thesis argument is based on this dependence. Regardless of the scheduler finding a promising group of users, the ultimate MU-MIMO gain relies on whether the precoder has an appropriate alignment with the actual multi-user channel geometry. When there are directional errors in the practical CSI, the RZF solution cannot be used to correctly correspond to the actual channel, and residual inter-user interference still exists. This interference influences the effective SINR, BLER, MCS distribution and throughput. Thus, the precoding step offers another fundamental route-after the scheduling step-by which enhanced CSI representation achieves system-level gain.

Taken together, the practical transmission chain used in this thesis consists of CSI-RS-based channel measurement, Type 2-oriented CSI representation, SUS+PF scheduling, and RZF precoding. This chain defines the concrete system context in which the value of improved CSI can be meaningfully evaluated. The next chapter explains how this practical operating framework was implemented in the OAI-based RAN digital twin.

2.3 5G NR CSI Acquisition Framework

2.3.1 CSI-RS and Channel Measurement

To acquire downlink CSI in 5G NR, it starts by transmitting a CSI reference signal to the gNB. The UE measures the received CSI-RS and estimates the downlink channel on the configured bandwidth and antenna ports. This estimation reliability is affected by such parameters as CSI-RS periodicity, number of ports, resource density, and effective SNR at UE receiver. Because the measured CSI will subsequently be used as the basis of adaptation of ranks, adaptation of beams, adaptation of links and MU-MIMO user pairs, the CSI-RS configuration has a direct impact on end-to-end performance of the system [2–4].

The UE does not usually provide a report on the entire channel matrix as it is. Rather, the measured channel is squeezed into useful feedback signals, with the most important ones being the Rank Indicator, the Precoding Matrix Indicator, and the Channel Quality Indicator. RI indicates the optimum transmission rank, PMI indicates a concisely represented best precoding direction or beam structure, and CQI indicates an estimate of effective channel quality to choose the modulation and coding. The indicators are the practical CSI interface between UE and gNB and thus they determine the information bottleneck in the downlink transmission chain [2, 4].

From the perspective of this thesis, the CSI acquisition process is important not only because it provides channel information, but because it determines how much of the actual channel structure can be preserved when that information reaches the transmitter. If this stage is inaccurate or too coarse, then all subsequent operations—including scheduling and precoding—must operate on distorted channel geometry.

2.3.2 Type 1 and Type 2 Codebook Based CSI Feedback

The 5G NR practical downlink CSI feedback is implemented based on the use of structured codebook-based reporting in contrast to the direct transmission of the whole channel matrix. Type 1 and Type 2 codebooks are two of the standardized feedback modes that are of practical significance, with different trade-offs between feedback overhead and channel-representation fidelity. Type 1 offers a comparatively small beam-oriented description, whereas Type 2 is more expressive as it enables a linear combination of several beams using extra coefficient information. Consequently, Type 1 is typically related to low feedback cost at the cost of low spatial resolution, whereas Type 2 gives enhanced channel fidelity at a significant rise in feedback overhead. It can be seen in 3GPP NR physical-layer procedures, which have been extensively debated in the literature regarding practical CSI feedback and codebook design [2, 24, 25].

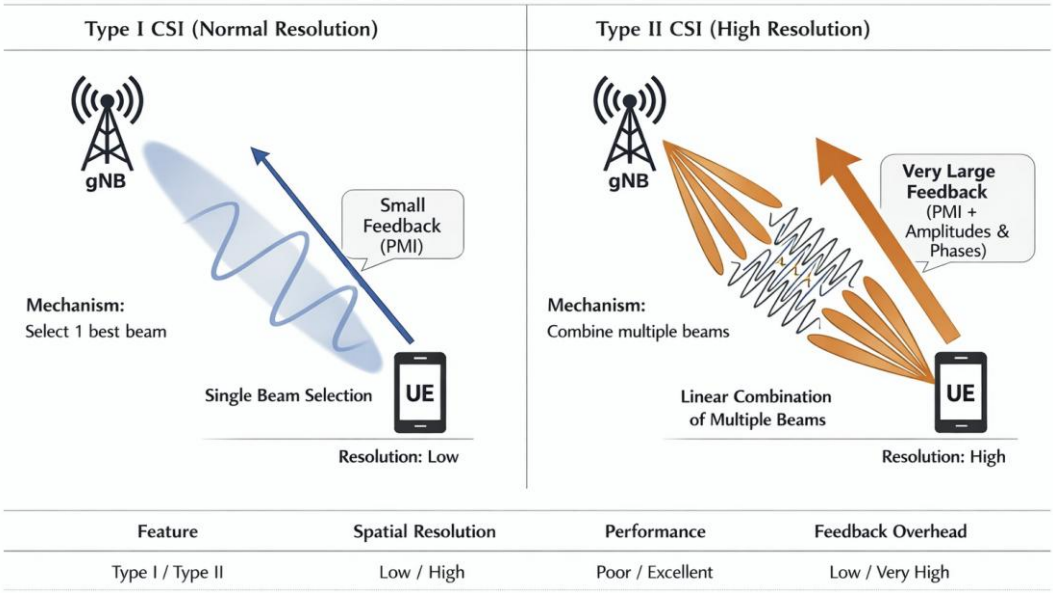

Figure 2.1 Comparison between Type I CSI and Type II CSI feedback

The primary distinction between the two schemes can be seen in Figure 2.1, indicating how the channel direction is depicted. As Type 1 CSI requires the UE to identify and report one dominant beam or one compact directional structure that best represents the measured channel. It is appealing since it minimizes the feedback payload and makes it easy to interpret the reported PMI at the transmitter. Its descriptive capability is weak in the case where the channel has more than two significant angular components or in the case where the precise directional discrimination is necessary. In this case, the transmitter will receive a rough approximation of the real channel structure.

In contrast, Type 2 CSI denotes a channel as a combination of several beams and their corresponding coefficients. Practically speaking, it implies that the UE does not only indicate which beam directions matter, but also the way these beams ought to be combined to give a better approximation of the actual channel. This enhanced representation provides better spatial resolutions and enables a transmitter to re-construct a channel direction, which can be much nearer to the real effective subspace of the measured channel. These kinds of representations may be especially valuable when there is no good approximation of a channel with a single dominant beam, which is frequently observed in realistic multipath propagation conditions.

The practical implication of this difference becomes more significant in MU-MIMO than in SU-MIMO. In SU-MIMO, a coarse beam description may still provide acceptable performance in

some operating conditions, especially when the channel is strongly directional and the transmission targets only one UE. In MU-MIMO, however, the transmitter must preserve not only each user's own effective channel direction but also the relative spatial relationship among multiple users. If the CSI representation is too coarse, then users that appear separable under the reported CSI may in fact be strongly interfering in the true channel, or valid multi-user opportunities may be missed because the reported channel directions do not preserve the true geometry accurately enough. For this reason, Type 2 is more suitable than Type 1 for the practical MU-MIMO transmission setting considered in this thesis.

At the same time, the higher fidelity of Type 2 comes with a substantial cost. Because it reports multiple beam components together with amplitude and phase information, its feedback overhead can become significantly larger than that of Type 1. Therefore, even though Type 2 provides a much stronger practical CSI operating point, it still does not remove the CSI bottleneck entirely. Instead, it defines a higher-quality but more expensive practical feedback mode. This observation is central to the present thesis. Since Type 2 is the more appropriate practical CSI representation for interference-sensitive MU-MIMO, the OAI-based RAN twin developed in Chapter 3 adopts a Type 2-oriented practical CSI path as the main conventional operating condition for multi-user transmission. The proposed AI/ML-based CSI encoder is then evaluated relative to this stronger practical operating point rather than only against a low-resolution Type 1 mode.

Chapter 3.

Design and Implementation of the OAI-Based RAN Digital Twin

This chapter presents the design and implementation of the OAI-based RAN digital twin developed in this work. The objective of the platform is not only to execute realistic end-to-end MIMO experiments but also to generate and extract long-term channel statistics for the proposed AI/ML model. Therefore, the platform plays a dual role. First, it acts as an execution platform in which CSI acquisition, feedback, scheduling, precoding, and throughput evaluation are performed under realistic channels. Second, it acts as a statistics extraction platform in which repeated channel realizations are used to estimate long-term descriptors such as covariance and PDP for the proposed statistics-conditioned encoder. This dual role, which directly follows the logic of the thesis, is also one of the platform's signature features.

3.1 Overview of the RAN Digital Twin

The suggested RAN digital twin was constructed based on the integration of three major subsystems, i.e., the 5G protocol stack with the OAI, the channel engine developed using Sionna and centralized control plane that orchestrates experiments. The experimental study presented in this thesis is carried out in a single-cell 4T4R setup, which allows investigating Type 1 SU-MIMO, Type 2 SU-MIMO, Type 2 MU-MIMO, and the proposed AI/ML-based CSI feedback scheme under a single framework. As illustrated in Figure 3.1, the OAI subsystem offers protocol-stack functionality to both the gNB and UE side and contains the core radio-access layer to enable end-to-end transmission, whereas the Sionna sub-system is tasked with generating channels and channel application in environments-friendly propagation conditions. The motivation behind this architectural decision is to integrate a realistic and modifiable 5G software stack with physically meaningful wireless channel generation. OpenAirInterface has been extensively employed as an open experimental tool to investigate 5G protocol-stack research and Sionna offer physical-layer simulation using GPU acceleration and differentiable ray-tracing-based propagation modeling.

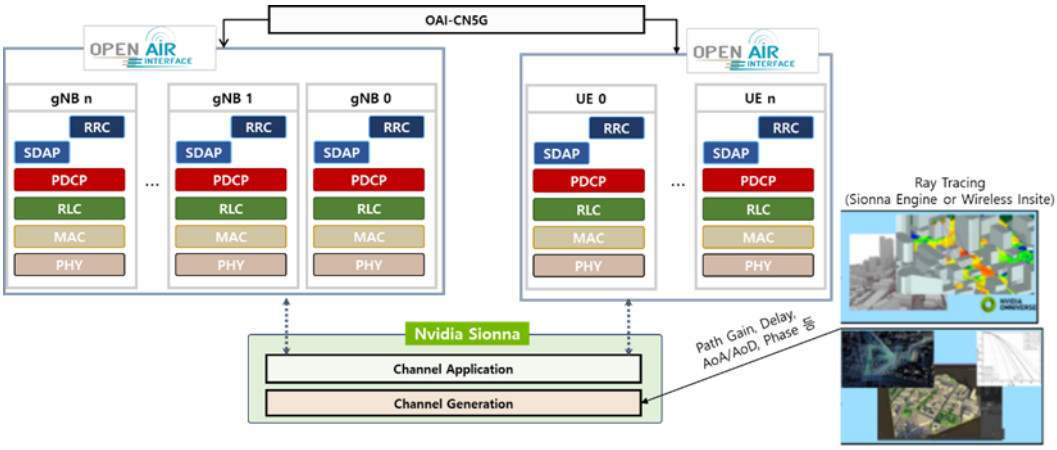

Figure 3.1 4T4R OAI-Sionna based RAN twin

In terms of functionality, the architecture can be viewed as consisting of two closely related roles. The first is an execution role where the OAI stack accomplishes the practical radio procedures like CSI-RS transmission, CSI measurement, CSI reporting, scheduling, precoding, and channel proxy-based data transmission. The other one is a statistics-extraction role, where the same physical situation can be used to obtain repeated channel realizations to estimate long-term descriptors including covariance-related and delay-domain statistics. Such descriptors are subsequently applied as side information to the planned AI/ML CSI encoder. The given coupling is an important methodological benefit of the current framework since it eliminates the usual discrepancy between the model-training environment and the system-level validation environment. More recent works on wireless digital twins have highlighted the importance of combined environment-conscious frameworks (to both assess networks and produce data) and the suggested framework is aligned in that overall sense but with particular emphasis on CSI feedback implementation in practical MIMO transmission.

A further significant design characteristic is that the multi-user channel geometry is physically meaningful. Since the Sionna engine produces channels of several UEs in the same scene, the relationship between user channels is defined by the common propagation environment, not by independent random variables. It is especially relevant to the MU-MIMO problem discussed in this thesis because scheduling stage user grouping depends directly on spatial compatibility between users. The quality of the inter-user geometry cannot be maintained in the underlying channel model, hence the practical performance of MU-MIMO scheduling and interference-aware precoding cannot be predicted with confidence. In contrast, the proposed architecture preserves such consistency across the generation of the environment to the execution of the protocol-stack, which allows it to

be used to investigate the end-to-end impact of the quality of the CSI representation on practical MIMO transmission.

3.2 Core Emulator: Centralized Configuration Management Server

One of the most significant challenges associated with experimentation in OAI systems is that the variables that control radio transmission, protocol behavior, channel emulation, and runtime execution tend to be dispersed among various configuration layers. In a typical workflow, radio-stack behavior-related settings, launch-time execution options, channel-model parameters, and service-level assumptions might be handled in different locations, making them challenging to modify together and raising their chances of inconsistency. The issue of such fragmentation is especially acute in the current work since the platform has to enable switching between the Type 1 SU-MIMO, Type 2 SU-MIMO, Type 2 MU-MIMO, and AI-enhanced CSI feedback modes repeatedly within a unified single-cell 4T4R experimental scenario. Due to this, the suggested RAN twin presents a specific Core Emulator that serves as a central configuration management server and creates a joint control plane over the entire experiment.

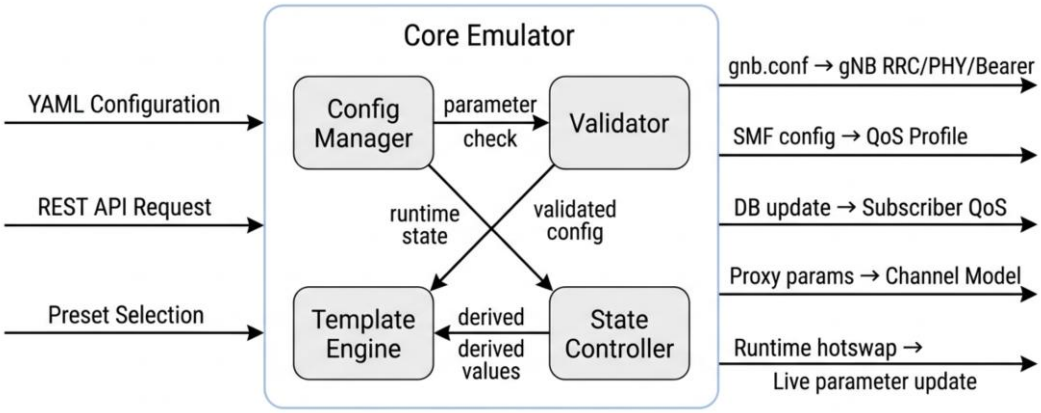

Figure 3.2 Functional architecture of the Core Emulator

The Core Emulator is based on four functional modules, i.e. a configuration manager, a validator, a template engine and a state controller, as illustrated in Figure 3.2. The configuration manager takes structured scenario descriptions and external control requests and translates them into a single internal representation of the experiment state. The validator subsequently checks

parameters and makes sure that radio, channel and service-layer assumptions are mutually consistent. Once validated, the template engine converts the internal experiment state into subsystem-specific configuration artifacts and the state controller retains derived runtime values and facilitates controlled updates at runtime. It is significant since the whole experiment can be controlled as a single integrated system state instead of an aggregate of unrelated low level states. Such specific behavior of these modules as checking parameters consistency, generation of derived-values, and managing runtime states is determined by the architecture of the designed Core Emulator.

The input-side of the Core Emulator is purposefully heterogeneous. It takes structured configuration input, programmatic control commands as well as predefined operating presets through a REST-style interface. The design has value in two aspects. To begin with, it enables reproducing of experiment specifications in that an entire radio scenario can be defined as a clear high-level state and not reconstructed using the disconnected implementation details. Secondly, it allows effective exploration of scenarios, whereby preset-based switching and API-driven changes enable systematic activation of differing modes of transmission and conditions of propagation. In the case of the current thesis, it is particularly critical because the identical platform needs to be utilized to compare various CSI feedback and transmission schemes with respect to different propagation conditions.

The core emulator has an output side as well. The control plane does not just store parameters but instead propagates validated configuration data to the major subsystems of the experimental platform. As it can be seen in the figure, those outputs are radio-access configuration of the gNB-side protocol stack, service-level QoS-related parameters of the core-network side, database-level subscriber updates when required, channel-model parameters of the proxy/channel engine, and selective runtime parameter updates of live execution. This design guarantees that all the main subsystems are running on the same interpretation of the active situation. This harmonious distribution of configuration is in line with the overall functionality of digital twins in communication systems whereby control, scenario definition and execution need to be kept up to date so that the subsequent experiments can be reproduced and explained.

One of the most significant roles of the Core Emulator is parameter validation. Many of the errors in complex MIMO experiments do not come about due to invalid individual values, but due to incompatible combinations of other values that are individually rational. An example is that a certain feedback mode may need a minimal antenna configuration, a particular scenario may imply a specific channel-model assumption, or a chosen service profile may be inconsistent with a desired transmission setup. The validator module thus does not only verify local parameter ranges but cross-layer consistency as well prior to conducting the experiment. The validation step is critical in this current study since the comparative findings in subsequent chapters rely upon separating the effect of CSI representation and transmission mode instead of unintentionally introducing hidden configuration mismatch.

A further essential ability is the capability of generating derived run-time values. In experimental radio practice, numerous operational parameters are not directly entered by the user, but they are inferred by more general parameters such as the antenna structure, the propagation mode, and the transmission setup. These calculated values are therefore used by the state controller to ensure uniformity between the radio stack, channel engine, and execution logic. It is especially significant in the current work as the platform is supposed to be able to support a working single-cell 4T4R system with several CSI feedback schemes and even minor internal discrepancies would make the final comparison less trustworthy. The Core Emulator makes sure that such dependencies are solved in a systematic way instead of manually.

The template engine has yet another methodological benefit. The activation of the experiment state can be verified and filled with derived values to generate subsystem-specific configuration outputs in an automated way. To put it simply, the experimental platform does not need to rewrite low-level artifacts each time it transitions between a high-level scenario description and operational subsystem settings. Essentially, the Core Emulator enables the hypothetical RAN twin to be controlled with a declarative scenario specification instead of a fragmented implementation-level intervention. It is particularly helpful when conducting a thesis-scale study and having to reproduce numerous scenarios and transmission-mode combinations across large number of experimental runs.

The Core Emulator also allows controlled runtime updates, besides the static configuration management. As shown in the figure, certain live parameter modifications are possible using a runtime hot-swap path. This functionality is significant since it is not necessary to restart the system in all variations of scenarios. E.g., some channel-side parameters or execution parameters can be changed under controlled experimentation without resetting the overall state of the experiment. This increases the efficacy of the digital twin and makes it more applicable to the iterative analysis of scenarios. It is also consistent with the overall notion of the digital twin where a virtualized system must enable reproducible configuration as well as controlled adaptation when running.

On the whole, the Core Emulator is an underlying component of the suggested RAN twin. It is not simply a convenience layer over the protocol stack, but the element that gives the whole framework experimental coherence. Through the receipt of structured scenarios inputs, checking parameter consistency, creating derived runtime states, rendering subsystem outputs, and propagation of updates across the radio, core-network, and channel-model parts, it guarantees that the next CSI acquisition, scheduling, precoding, and performance evaluation steps are executed within the same unified experimental state. Such centralized control architecture is thus critical to the reproducibility and interpretability of the end-to-end MIMO experiments described in the rest of the thesis.

3.3 Sionna Channel Proxy: GPU-Accelerated MIMO Channel Emulation

A central requirement of the proposed RAN digital twin is that the protocol stack must be able to communicate to a realistic wireless channel without disrupting the multi-user nature of the communication network. In order to do this, the platform also has a channel-proxy layer, which interfaces between the OAI-based radio stack and the Sionna-based channel engine using IQ-samples. Within this architecture, the transmitted baseband samples produced by the radio stack are sent to the channel emulator, where they have been modified based on environmental MIMO channel coefficients, and then sent back to the appropriate receiver side. With this design, the protocol stack can function in a manner as though it was coupled to a real life over-the-air propagation environment but all execution is done in a controlled software-defined environment.

The primary task of the channel proxy is not merely to add fading to a single-user waveform. Instead, it has to maintain the architecture of a multi-user wireless network where several UEs share the same radio environment. In the downlink, the channel formed by the transmitted signal produced by the gNB is duplicated to many different UE-specific transmission channels such that every UE observes an entirely different version of the channel depending on its position, geometric shape, and propagation circumstances. In the uplink, waveforms sent by many UEs are separately filtered using their respective channels and then combined by the gNB receiver. Such a bidirectional arrangement is necessary as the thesis examines realistic MU-MIMO operation, where relative geometry between the users plays a significant role in determining the performance of scheduling and precoding. An independent user channel emulator without keeping track of the common environmental context of users could not be a useful reference point to evaluate MU-MIMO.

The combination with Sionna has two significant benefits. Firstly, it makes it possible to use the process of generating and applying MIMO channel coefficients using a GPU, which is required due to the fact that running end-to-end experiments repeatedly and generating large datasets is computationally intensive. Secondly, it also enables the platform to take advantage of channel models offered by Sionna and, as required, propagation based on ray tracing. The implication here is that the channel coefficients produced are not just random fading samples but realization of channels with knowledge of the environment whose structure is related to the geometry of the city, propagation conditions, and assumptions of mobility. Considering the present thesis, this reality is especially vital since the scheduler and precoders will subsequently rely on slight variations in user channel geometry that should be maintained accurately.

One of the other key roles played by the channel-proxy layer is its ability to act as the interface whereby the long-term channel statistics are extracted. Because the identical scenario can be repeated many times in the controlled conditions, the same channel engine will produce several realizations of the same user or environment class. The repeated samples are subsequently employed

to approximate long-term descriptors like the covariance structure and the distribution of energy in the delay-domain. In such a manner, the channel-proxy subsystem is not merely an executing system but also the part of the data-generation pipeline in the proposed AI/ML CSI encoder. This dual functionality is a characteristic of the digital twin in this thesis: the same environment that operates live MU-MIMO transmission also acts as a source of side information that constrains the learned feedback model.

In terms of implementation, the channel proxy has been developed with the view of execution efficiency. Because the digital twin needs to be capable of performing repetitive sweep of scenarios and multi-UE experiments, the cost of channel creation and usage should be kept at a minimum to allow the realistic non-real-time experimentation to occur. To do it, the platform uses the execution path based on GPUs and organises channel generation in a pipeline-friendly fashion. Even though real-time operation is not what the current thesis aims at, steady slot-level operation is needed to ensure that the long traces of experiments are gathered and compared between situations. The ensuing design is a reasonable trade-off between physical realism and execution feasibility, which is adequate to perform end-to-end evaluation performed later in the thesis.

One of the most useful properties of such proxy architecture is that it maintains consistency between the environment in which the data were generated and the environment in which the system was evaluated. Training channels are created in one framework and system-level validation is conducted in another, frequently due to mismatched assumptions, in much of the AI-for-wireless literature. By contrast, the channel engine of the current thesis is the same across both tasks. Consequently, the statistical descriptors provided to the proposed encoder come out of the same scenario family that will eventually control the actual execution environment. The consistency adds credibility to the performance improvement reported in subsequent chapters since the model is not being tested in an environment separate from where its conditioning information was acquired.

3.4 MU-MIMO Scheduling and Precoding Implementation

3.4.1 Type 2 PMI-Based Precoding

A practical evaluation of MU-MIMO in the proposed RAN twin requires that the CSI reporting path provide a sufficiently informative channel-direction representation to the gNB. For this reason, a Type 2-oriented PMI processing path was implemented so that the reported CSI could be used not only as a measurement result, but also as a functional input to multi-user transmission decisions. This implementation step is particularly important because the goal of this work is to compare actual limited feedback operations with statistically conditional AI/ML CSI encoders in the actual protocol stack, rather than just comparing them with idealized full CSI settings at the offline channel reconstruction level. In 5G NR, Type 2 CSI provides a richer channel description than Type 1 by

representing the channel through a linear combination of multiple beams and associated coefficients, and is therefore more suitable for interference-sensitive MIMO transmission when higher channel-direction fidelity is required.

In the implemented system, the UE-side CSI estimation path was extended so that the reported directional information reflects the spatial structure of the target 4×4 MIMO configuration in a practically meaningful manner. The reported PMI is then interpreted at the gNB side as a compact channel-direction descriptor and is used to regenerate a transmitter-side channel approximation. In this sense, the Type 2 PMI is not treated merely as a standardized feedback quantity but as a runtime variable that directly affects the downlink multi-user transmission chain. The same regeneration framework was also constructed for Type 1 PMI so that the quality of the two practical CSI representations could be compared under a common evaluation methodology.

The verification of this implementation was carried out through PMI-based channel regeneration. More specifically, for both Type 1 and Type 2 CSI, a reconstructed channel $\hat{\mathrm{H}}$, referred to here as a quantized channel, was regenerated from the reported PMI information. This regenerated channel was then compared with the corresponding true channel H obtained before practical CSI quantization. In order to quantify the fidelity of the regenerated channel, the normalized correlation between the true channel and the quantized channel was evaluated. Let h and $\hat{h}$ denote vectorized forms of the true and reconstructed channels, respectively. The comparison metric can be written as

$$\rho = \frac{\left|h^H \hat{h}\right|}{\|h\|\|\hat{h}\|}, \tag{3.1}$$

a larger value of ρ means that the original channel and the recreated channel representation are more closely aligned.

By this measure, the channel quality of Type 1 and Type 2 based regeneration was studied as a dependence on SNR. The aim of this assessment was to find out if the adopted Type 2 CSI path does really maintain the channel structure more precisely compared to its Type 1 counterpart in more favorable measuring conditions. It is apparent in Figure 3.3 that the mean normalized correlation between the true channel and the regenerated channel grows with SNR in both methods, as expected because an increased SNR results in a more reliable measurement and reporting of the CSI. Nevertheless, the increase in gain of Type 2 can be seen to become more pronounced as the SNR rises. Although the two methods have nearly the same behavior at low SNRs, the Type 2 method has significantly higher correlation at moderate-to-high SNRs, which suggests that the more extensive CSI structure of Type 2 is effectively converted into a more precise reconstructed channel. This pattern is entirely in line with the anticipated benefit of Type 2 CSI feedback described in the 5G NR literature.

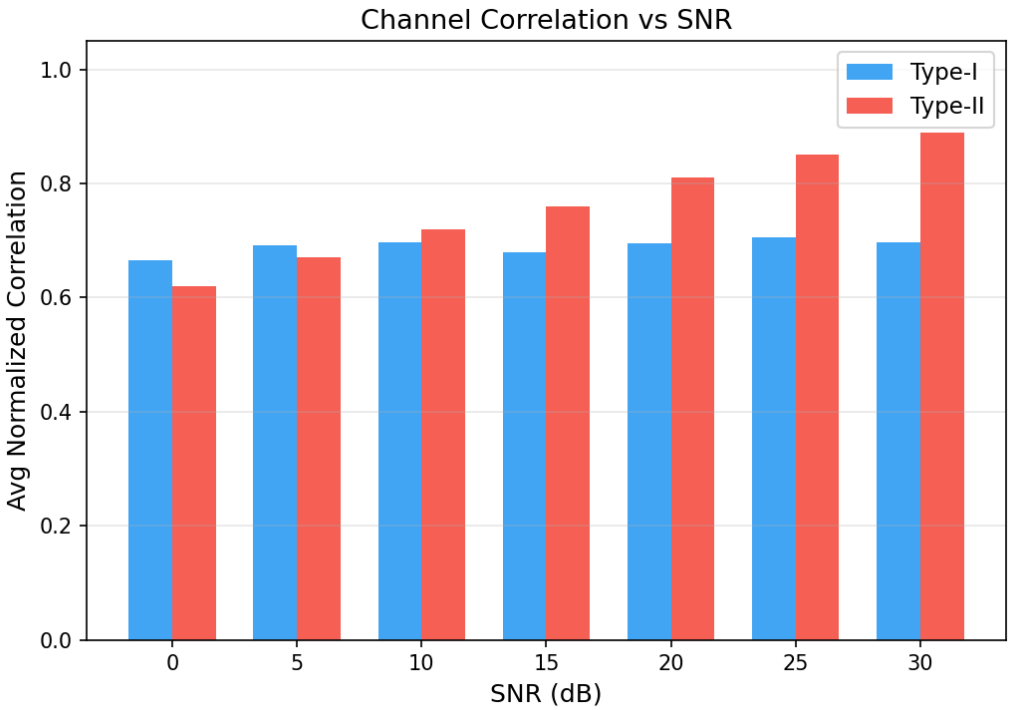

Figure 3.3 Average normalized correlation between the true channel and the regenerated quantized channel obtained from Type-I and Type-II PMI

This finding can be used as the fundamental validation of the developed Type 2 PMI route. It is significant to note that it is not just that Type 2 is theoretically more expressive than Type 1, but that the suggested RAN twin implementation effectively recreates this anticipated benefit using a quantifiable channel-regeneration measure. To sum up, the introduced CSI reporting and regeneration route maintains the relative fidelity difference between Type 1 and Type 2 both in a quantitative sense and in a physically plausible way. It is necessary in the rest of the thesis since further MU-MIMO scheduling and precoding experiments depend on the practical CSI path being meaningful and not nominally present.

3.4.2 Scheduling Algorithm Implementation

The scheduling architecture in the suggested RAN twin was deployed because MU-MIMO user selection is done internally within the real gNB scheduling path as opposed to externally outside the

execution loop. The primary aim of this implementation was the assurance that user-grouping choices are based on the similar practical CSI representation used by the protocol stack and thus realism of the end-to-end evaluation is maintained. To put it another way, the scheduler was built to work on the CSI information that is reported and reconstructed instead of on the hidden perfect channel knowledge. This design is in line with the practical reasons behind semi-orthogonal user selection in limited-feedback MU-MIMO systems where user grouping needs to be established based on transmitter-available CSI instead of using the perfect full channel observations.

The scheduler determines at each scheduling interval which UEs should be considered as candidates to receive downlink transmissions based on the prevailing traffic conditions and resource status. Following this preliminary filtering of candidates, the MU-MIMO extension decides if it is possible to serve more than two UEs simultaneously. In this grouping step, the practical directional information based on the CSI reporting is utilized and relative compatibility between candidate users is checked. The condition of spatial compatibility and the fairness-aware utility are considered before the last selection of users is made. Therefore, MU-MIMO grouping is included in the runtime scheduling sequence instead of being an analytical operation that follows the events.

One of the significant implementation challenges was that traditional transmission flow was primarily aimed at single-user-at-a-time scheduling behavior. In order to facilitate the desired MU-MIMO operation, it was necessary to extend the scheduling logic to enable joint evaluation of more than two UEs during the same interval. This necessitated the system to keep CSI based transmission descriptors by user basis and comparison prior to final resource allocation. When a multi-user compatible group has been determined, the chosen UE set is then sent to subsequent transmission steps in a uniform way such that scheduling, interpreting channels and construction of precoders all concern the same set of users.

The second key design consideration was that the scheduler must not be insensitive to the quality of the practical CSI representation. Because the overall aim of the thesis is to examine how the accuracy of the CSI influences end-to-end MU-MIMO performance, the scheduler was deliberately restricted to the CSI information which could be present in a practical limited-feedback system. This implies that schedule results will depend directly on the quality of CSI. In case the reported channel directions are inaccurate, user grouping is no longer reliable; in case the directional information is accurate, the scheduler can make better use of the MU-MIMO opportunities. It is a valuable feature to consider when evaluating it subsequently.

Basic verification of the implemented scheduling path was carried out through runtime MU-MIMO grouping logs obtained under an eight-UE setting. As shown in Figure 3.4, the scheduler selects four UEs out of eight candidates in a given scheduling interval and assigns them to distinct DMRS ports, resulting in four-layer spatial multiplexing. At the same time, the remaining UEs are explicitly rejected by the SUS procedure, together with their measured correlation values. In the example shown in the figure, users with high correlation values are rejected from the same MU-

MIMO group, while users with sufficiently lower mutual correlation are retained for transmission. Such behavior is precisely the one predicted by a semi-orthogonality-based scheduling model: the algorithm maintains users who can be transmitted simultaneously and removes the ones whose channels are similar enough so that suppressing multi-user interference becomes challenging.

```
┌─ MU-MIMO Group #1  [frame 155, slot 2]  4 UEs selected from 8
│  UE0 (RNTI=0x0c17)  DMRS port=0  RB 0+48  MCS=6   1L  pm=147 (ZF)
│  UE5 (RNTI=0x72f6)  DMRS port=1  RB 0+48  MCS=8   1L  pm=170 (ZF)
│  UE3 (RNTI=0xee0d)  DMRS port=2  RB 0+48  MCS=2   1L  pm=151 (ZF)
│  UE7 (RNTI=0xc8e3)  DMRS port=3  RB 0+48  MCS=5   1L  pm=185 (ZF)
│  [SUS] rejected: UE1(corr=0.961), UE2(corr=0.974), UE4(corr=0.952), UE6(corr=0.968)
└─ 4/8 UEs selected, 4 DMRS ports → 4-layer spatial multiplexing

┌─ MU-MIMO Group #2  [frame 162, slot 17]  4 UEs selected from 8
│  UE1 (RNTI=0xaa8e)  DMRS port=0  RB 0+48  MCS=8   1L  pm=138 (ZF)
│  UE4 (RNTI=0x3b21)  DMRS port=1  RB 0+48  MCS=6   1L  pm=162 (ZF)
│  UE2 (RNTI=0xda59)  DMRS port=2  RB 0+48  MCS=2   1L  pm=136 (ZF)
│  UE6 (RNTI=0x519a)  DMRS port=3  RB 0+48  MCS=4   1L  pm=178 (ZF)
│  [SUS] rejected: UE0(corr=0.958), UE3(corr=0.981), UE5(corr=0.965), UE7(corr=0.943)
└─ 4/8 UEs selected, 4 DMRS ports → 4-layer spatial multiplexing
```

Figure 3.4 Runtime verification of the implemented MU-MIMO scheduling path in an eight-UE setting

This verification has two aspects. Firstly, it indicates that the scheduler does not get stuck in a single-user model but actually creates multi-user clusters at run-time. Secondly, it demonstrates that the SUS-based rejection mechanism is employed effectively in the execution chain and the grouping decision is based on user-correlation data, not on a fixed ordering of UEs. To put it another way, the deployed scheduler is not simply able to report several selected users, but it is conducting the desired geometry-aware selection operation in the real transmission channel.

Another significant fact that can be found in the runtime logs is that there is a consistency of the scheduler between user selection and transmission layer allocation. The chosen users are connected to different DMRS ports and are ready to transmit simultaneously at multi-layers, which implies that the scheduling result can be immediately used by the next transmission process. It confirms that the scheduling operation is incorporated in the downstream pre-coding and transmission control instead of being a separate decision unit.

Based on the general thesis, this verification indicates that the applied scheduling step offers an actual system level pathway wherein the quality of CSI may affect the end-to-end performance.

As user clustering is based on practical CSI, better CSI representation may result in more dependable MU-MIMO cluster generation. Hence, the scheduling phase becomes one of the essential components whereby the suggested CSI feedback improvement would be subsequently converted into throughput improvement.

3.4.3 Precoding Algorithm Implementation

When the user set has been chosen by the scheduler, the RAN twin will calculate the downlink precoder through a regularized zero-forcing structure. The aim of the implementation was to make sure that the practical CSI information present in the gNB can be converted into a good multi-user channel representation and subsequently applied directly as runtime downlink precoding. With this approach, the precoding phase will be included as a real execution branch of the twin and not just a conceptual reference framework.

The proposed precoding path starts with forming a useful user-channel matrix based on the chosen UE set. Based on the practical channel direction data reported by the CSI reporting chain, a transmission side directional representation is formed to map the channel direction data of each scheduled UE. Such per-user directional components are subsequently grouped into a multi-user channel matrix that is fed into the regularized inversion stage. The obtained result is a precoding matrix that will be applied to separate the data streams of the desired UEs prior to their transmission. Since this procedure is determined exclusively by practical CSI instead of ideal full channel information, the final precoder represents the real information status in which the system is functioning.

One of the significant challenges to implement was that the precoding process should be made compatible with the scheduling decision. Because the scheduler has already decided the set of users to be co-scheduled, the precoder needs to be calculated based on the chosen subset and needs to interpret the CSI in the same way as the scheduler. In case both the scheduler and the precoder would have been based on non-uniform channel models, the resulting MU-MIMO performance would be impossible to interpret. The implementation was designed to make sure that the chosen UE set and its practical CSI descriptors are passed to the precoding stage simultaneously to allow deriving the RZF matrix based on a coherent multi-user channel model.

Another crucial point is that the developed precoder should be able to work well in the presence of imperfect CSI. Theoretically, strict zero forcing might have been applied in a simulated environment where precise channel knowledge was available to wipe out all the inter-user interference. But in the suggested twin, the CSI accessible to the transmitter is bounded by the errors of practical reporting and reconstruction. Hence, a regularized zero-forcing structure is preferable since it does not compromise the main interference-suppression concept but is numerically more

robust and less sensitive to CSI distortion. This is why RZF was selected as the practical implementation target of this work rather than a pure zero-forcing design.

From the perspective of the thesis, the precoding implementation provides the second major system-level pathway through which CSI feedback quality affects performance. The scheduler determines whether multiple users should be served together, but the precoder determines whether they can be spatially separated during transmission. Therefore, once the scheduling implementation and the precoding implementation are both integrated into the RAN twin, the platform becomes capable of showing how improved CSI representation affects MU-MIMO performance through both user grouping and interference suppression.

3.5 RAN Scenario Configuration and Initial Performance Verification

In order to test whether the suggested RAN twin generates meaningful results in terms of physical transmission behavior and consistency in wireless conditions, three typical urban propagation models were created and compared to each other to conduct the preliminary comparative assessment of feasible MIMO transmission schemes. One aim of this step was to ensure that the combined OAI-Sionna framework would produce the right performance ranking across traditional transmission configurations prior to the introduction of AI/ML-based CSI feedback techniques. Due to this fact, the first round of performance testing in this part is limited to three practical schemes only, which are Type 1 SU-MIMO, Type 2 SU-MIMO and Type 2 MU-MIMO in a single-cell setting of 4T4R at 40 MHz in FR1 Band 78. Selected scenarios are UMi-LOS, UMa-NLOS and UMa-LOS, which represent different propagation conditions described in the framework of the 3GPP channel model, which are commonly employed to characterize actual urban radio conditions.

The justification behind these three scenarios is their various requirements on CSI representation and multi-user transmission. UMi-LOS is the depiction of urban micro-cell environment with high line-of-sight part and comparatively directional structure. UMa-NLOS is the description of urban macro-cell environment with reflected and scattered propagation dominating, which leads to more diverse multipath structure and wider channel support. UMa-LOS refers to a macro-cell environment with a prevalent LOS path and a large scale user spread and thus a large scale urban LOS condition has practical relevance. The differences between these scenarios are significant since CSI fidelity and MU-MIMO gain strongly rely on propagation geometry, user separability, and the complexity of the effective channel perceived by the transmitter.

The same general procedure was used in all scenarios. The first step involved setting up the environment-aware channel engine based on the desired propagation condition. Then, the end-to-end transmission chain was implemented in three operational conditions, which were Type 1 SU-

MIMO, Type 2 SU-MIMO, and Type 2 MU-MIMO. The metrics that were compared were block error rate, average MCS index and cell downlink throughput as a function of the number of active UEs. As all three modes were assessed in one unified setup, the observed differences are not the result of individual changes in configurations but rather are the cumulative effect of both CSI representation and transmission mode. Methodologically, this type of preliminary check is significant since it ensures that the twin is enough sensitive to realistic quality of CSI and MU-MIMO operation prior to more sophisticated CSI feedback models being introduced.

The results for the UMi-LOS scenario are shown in Figure 3.5. In the BLER result, Type 1 SU-MIMO exhibits the highest error rate over the entire UE range, remaining around 4.2–4.3%, while Type 2 SU-MIMO stays close to 4.0%. Type 2 MU-MIMO provides the lowest BLER when the number of active UEs is small, starting from about 3.5% at one UE and increasing gradually as the UE count grows. However, when the number of active UEs reaches eight and sixteen, the BLER of Type 2 MU-MIMO becomes slightly higher than that of Type 2 SU-MIMO. A similar tendency is observed in the average MCS index. Type 2 MU-MIMO gives the highest MCS in the low-UE region, but at eight and sixteen UEs, Type 2 SU-MIMO shows slightly better MCS values than Type 2 MU-MIMO.

In contrast, the trend of the cell downlink throughput is different. The Type 1 SU-MIMO is constantly the lowest one with a decrease of about 33 Mbps at one UE and about 15 Mbps at sixteen UEs. Type 2 SU-MIMO is initially about 70 Mbps and then decreases gradually with increasing number of UEs. Type 2 MU-MIMO is lower than Type 2 SU-MIMO initially but when only one UE is active, it grows to about 67 Mbps, 82-83 Mbps, 4 and 8 UEs respectively, and levels off at about 60 Mbps at 16 UEs. Thus, despite the fact that Type 2 SU-MIMO is slightly superior to Type 2 MU-MIMO in the case of BLER and average MCS at higher UE counts, Type 2 MU-MIMO remains the best in terms of aggregate cell throughput in the multi-user region.

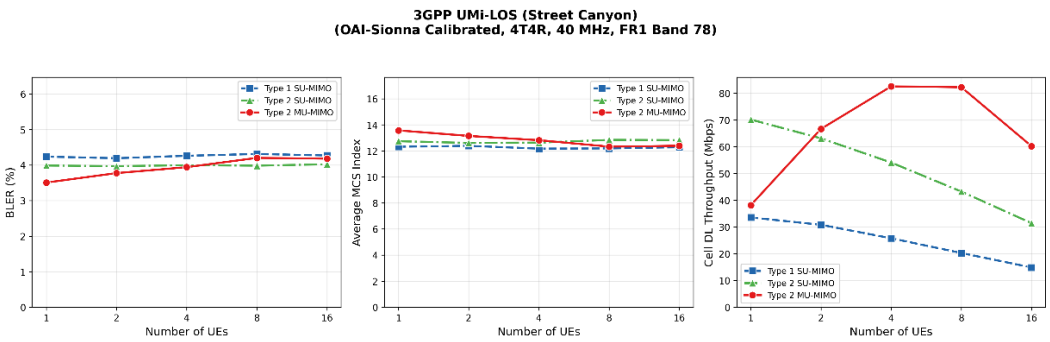

Figure 3.5 Initial performance verification in the UMi-LOS scenario

The behavior has physical significance and indicates that the relative benefit of SU-MIMO and MU-MIMO is based on what performance measure is used. With one UE, Type 2 SU-MIMO is outperforming since MU-MIMO cannot take advantage of any spatial multiplexing benefits unless there are several active users at the same time. With two or four UEs, the Type 2 MU-MIMO throughput is now clearly the best because the scheduler is able to identify beneficial user groups and utilize multi-user transmission. But when the number of active UEs grows, e.g., eight or sixteen, it becomes harder to group users in the UMi-LOS environment. Because LOS-dominated streets canyon channels are likely to have high levels of directional focus, most users would seem to have a high degree of alignment in space. Therefore, the scheduler can still create MU-MIMO groups to increase cell throughput, yet the chosen users will not be as orthogonal as in the low-UE area. It causes residual inter-user interference after precoding that might reduce BLER and average MCS minimally despite the overall cell throughput being greater.

Another reasonable explanation could also be that MU-MIMO optimizes total throughput not necessarily the performance of individual transmission links. All the resources are used by all UEs one at a time in Type 2 SU-MIMO, which may cause slightly higher per-link reliability and choice of MCS. The available spatial degrees of freedom in Type 2 MU-MIMO have to be divided amongst multiple simultaneously served UEs. In a 4T4R environment, this implies that the effective multi-user separation ability is limited, particularly when the user channels are not very diverse, in a LOS-dominant environment. Hence, Type 2 SU-MIMO may exhibit somewhat lower BLER and MCS at a higher UE count, whereas Type 2 MU-MIMO would continue to demonstrate greater aggregate cell throughput.

This finding is important as it demonstrates that the suggested RAN twin represents a viable compromise between link quality and cell-level multiplexing gain. The behavior observed in the UMi-LOS case is not merely a flat out dominance of a single transmission mode by another one but rather it is shown that MU-MIMO remains the most efficient mode of providing maximum cell throughput even at conditions where_SU-MIMO starts to be slightly more efficient in BLER and MCS with increasing numbers of users.

The results of UMa-NLOS are presented in Figure 3.6. As it can be seen, in this case, there is an even greater disparity between the three transmission schemes. Type 1 SU-MIMO has the highest BLER with a significant margin, which remains relatively constant at approximately 26-29 percent across all the considered UE numbers. The reduction in this level is significantly lower in Type 2 SU-MIMO, at about 15-18, and Type 2 MU-MIMO is the lowest BLER with the BLER approaching 9 percent at one UE and slowly rising to about 14-15 percent with sixteen UEs. These findings prove that in a richer NLOS propagation, the enhanced CSI representation of Type 2 and the multi-user transmission capability are very useful to support the reliability of links.

The average MCS index follows the same ordering. Type 1 SU-MIMO remains lowest, around 2.4–2.6. Type 2 SU-MIMO improves the MCS to around 4.1–4.4, and Type 2 MU-MIMO achieves the highest values, roughly between 4.7 and 6.0. This indicates that the richer practical CSI and the implemented MU-MIMO chain allow the system to maintain considerably better effective transmission quality under NLOS macro-cell propagation.

The throughput result again shows a clear crossover between Type 2 SU-MIMO and Type 2 MU-MIMO. Type 1 SU-MIMO is always the weakest, decreasing from about 8 Mbps to below 4 Mbps. Type 2 SU-MIMO starts from about 25 Mbps at one UE and then decreases gradually as the user count increases. Type 2 MU-MIMO starts below Type 2 SU-MIMO at one UE, rises sharply to about 27–28 Mbps at two UEs, remains the strongest at four and eight UEs, and then decreases to around 18–19 Mbps at sixteen UEs.

Therefore, the same trend is also repeated in UMa-NLOS: Type 2 SU-MIMO performs better than Type 2 MU-MIMO at a single UE case, but Type 2 MU-MIMO becomes higher-performing with multiple active UEs. It is once again because multi-user gain can be achieved only with more than one candidate user to service at the same time. With one UE active, the MU-MIMO mode does not have any advantage of multiplexing. But as the number of active UEs grows, the scheduler gets additional freedom to make more beneficial user groups and the MU-MIMO throughput benefit becomes more apparent. The observation that the throughput achieves its maximum under two UEs in this context and then decreases steadily implies that in UMa-NLOS, the multi-user gain exists but is more prone to the increased number of users and complexity of channels than in UMi-LOS.

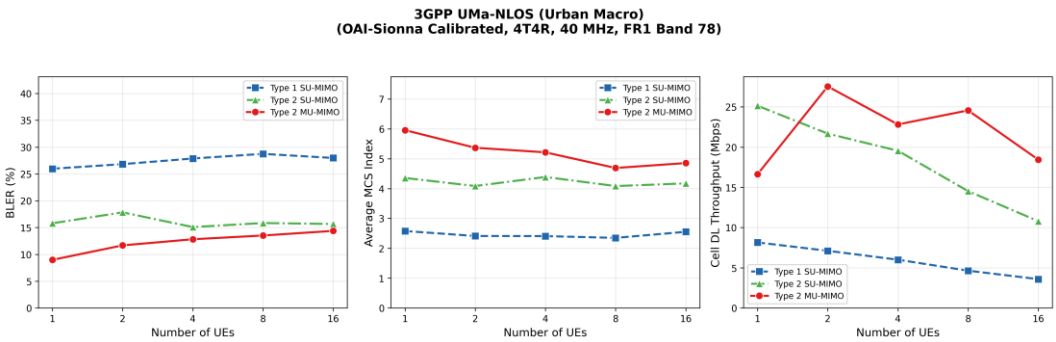

Figure 3.6 Initial performance verification in the UMa-NLOS scenario

The findings related to the UMa-LOS scenario are presented in Figure 3.7. With regard to the BLER outcomes, Type 1 SU-MIMO has the highest error rate across all UE ranges at around 8.7%-

9.4%. Type 2 SU-MIMO lowers the BLER significantly by about 6.0-6.9% but Type 2 MU-MIMO offers the least BLER in low to medium UE area with around 4.9% at one UE and it increases slowly as more UEs become active. As the UE count increases, the difference in BLER between Type 2 SU-MIMO and Type 2 MU-MIMO is extremely small and at sixteen UEs the two modes exhibit almost the same reliability rate. This shows that the effective MU-MIMO benefit in the given case is the most intense when the scheduler does not have such a small user variety that it cannot make favorable groups without significantly reducing the quality per link.

A trend of the same kind can be seen in the average MCS index. Type 1 SU-MIMO is the lowest, with an average of 5.2 - 5.4, and Type 2 SU-MIMO is better with an average of 7.1 - 7.3. Type 2 MU-MIMO has the highest values of MCS across the whole range of UEs with maximum values of approximately 9.0 with one UE active and decreasing progressively down to about 7.4 with growing numbers of UEs. Thus, unlike the UMi-LOS situation, the UMa-LOS outcomes are not characterized by any reversal in MCS sequencing at large UE loads. However, Type 2 MU-MIMO is slightly better than Type 2 SU-MIMO, but the gap is narrowing with the system moving into the large-UE region.

The most significant overall trend in the cell downlink throughput is also revealed. The weakest is type 1 SU-MIMO that is reduced by approximately 14.5 Mbps with one UE and less than 7 Mbps with 16 UEs. Type 2 SU-MIMO has an initial value of around 38 Mbps with a steady decline with an increase in UE count until it reaches approximately 17-18 Mbps with 16 UEs. Type 2 MU-MIMO has a different behavior, it is lower than type 2 SU-MIMO when there exists only one UE, but it is higher starting with two UEs. More specifically, it goes up to about 41 Mbps at two UEs, up to about 49 Mbps at four and eight UEs and then drops back to about 36 Mbps at sixteen UEs.

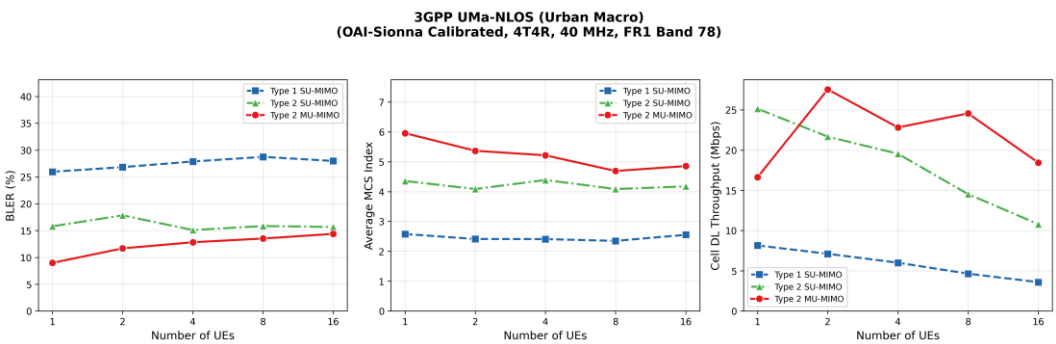

Figure 3.7 Initial performance verification in the UMa-LOS scenario

The action is physically significant. In case of one active UE, the MU-MIMO mode is not able to utilize any multi-user multiplexing gain, and thus Type 2 SU-MIMO attains a higher throughput by directing all transmission resources to a single user. But with the number of active UEs being more than one, Type 2 MU-MIMO can take advantage of concurrent transmission and hence delivers the maximum total cell throughput. The peak throughput at four-eight UEs indicates that it is within this range where the best tradeoff between user diversity and practical user separability is obtained under UMa-LOS conditions.

A significant note is that although Type 2 MU-MIMO is still the better choice in terms of throughput and average MCS, its BLER slowly tends to Type 2 SU-MIMO as the number of active UEs grows. One possible reason is that with a 4T4R setup, the number of spatial degrees of freedom is finite. The dominant LOS structure in the UMa-LOS case also leads to some level of directionality similarity between users, particularly since the scheduler has a wider selection of active-users. Thus, when the number of UEs is large enough, multi-user grouping can involve users with not so ideal channel separation, which could somewhat reduce the reliability benefit of MU-MIMO despite the overall throughput benefit being evident. To put it differently, the UMa-LOS findings indicate that the suggested RAN twin reflects a practical trade-off between per-link reliability and cell-level multiplexing gain: Type 2 MU-MIMO remains the most effective in terms of cell throughput, whereas the BLER performance compared to Type 2 SU-MIMO decreases at the high-UE location.

When considered together, the results in Fig. 3.5-3.7 reveal a number of consistent trends. Firstly, BLER, average MCS, and cell throughput are all better with Type 2 SU-MIMO than Type 1 SU-MIMO. It is evident that the more complex representation of CSI in Type 2 is accurately converted into actual gains in the RAN twin. Secondly, the best overall performance is achieved by Type 2 MU-MIMO when the number of active UEs is large enough to confirm that the deployed scheduling and precoding chain can utilize multi-user spatial multiplexing gain in realistic propagation conditions.

Concurrently, the findings indicate that in the case of extremely low active UE numbers, including the case of one UE, Type 2 SU-MIMO can be better than Type 2 MU-MIMO. It is possible because MU-MIMO needs enough active users at the same time to achieve spatial multiplexing gain. The MU-MIMO mode cannot take advantage of its biggest advantage when only one UE is active, since the system operates similar to a single-user system. Also, a practical MU-MIMO transmission could have additional overhead or an unfavorable resource allocation structure compared to a single-user transmission mode in this limit case. With growing number of active UEs, nonetheless, such constraints are overcome by the advantage of multi-user transmission, and Type 2 MU-MIMO stands out as superior.

A further possible reason is the state of user separability in practical CSI. MU-MIMO also needs not just several active UEs but also adequate channel direction differences between them. With a low number of UEs, the scheduler will have fewer options and might not discover the most

favorable user combinations. Increasing the number of users provides the scheduler with greater diversity and allows it to choose the combination of users with the best spatial compatibility which makes the benefit of MU-MIMO more noticeable. It is the reason why the throughput benefit of Type 2 MU-MIMO reaches its peak at the intermediate-to-larger UE-count region instead of the one-UE scenario.

Overall, these results provide an important initial validation of the proposed RAN twin. The platform reproduces the expected improvement from Type 1 to Type 2 CSI in single-user transmission, and it also reproduces the expected transition from single-user advantage at extremely small user counts to multi-user advantage when the system has enough active UEs to exploit spatial multiplexing. Therefore, the twin is sufficiently realistic and sensitive to practical CSI quality to serve as the experimental foundation for the learned CSI feedback evaluation carried out in the later chapters.

Chapter 4.

Design and Performance Evaluation of the Statistics-Conditioned AI/ML CSI Encoder

This chapter presents the proposed AI/ML-based CSI feedback framework of this thesis. The overall logic of the chapter is as follows. First, the limitations of blind CSI autoencoding are formalized in the context of realistic MU-MIMO channels generated from the RAN digital twin. Second, a baseline CsiNet-style autoencoder is introduced as the reference neural CSI compression architecture. Third, the proposed statistics-conditioned CSI encoder is developed by explicitly incorporating long-term channel statistics, extracted from the digital twin, into the encoding and decoding process. In particular, the proposed framework exploits channel covariance and power delay profile information as side information that changes slowly over time and can therefore be reported or maintained with a much lower average overhead than instantaneous CSI. Finally, the chapter presents the offline evaluation framework and discusses how the proposed method is compared against both the CsiNet baseline and conventional Type 2 codebook-based feedback under a fair bit-budget setting.

One of the main hypotheses of this chapter is that instantaneous CSI cannot be compressed alone in realistic urban propagation conditions. The encoder may be significantly more efficient by conditioning on such statistical information because the channel realizations are sampled by an environment-dependent distribution and the digital twin gives access to long-term descriptors of that distribution. In this regard, the suggested approach does not just enhance neural compression via a bigger backbone or deeper architecture. Instead, it alters the formulation of the CSI feedback problem in general: it is no longer about reconstructing the full channel blindly, but to reconstruct the instantaneous channel with a short description of its long-term environment-dependent structure.

4.1 Problem Definition and Motivation

Current deep-learning-based CSI compression algorithms such as CsiNet and most of its variants are able to compress the instantaneous channel matrix H without an explicit understanding of the environment in which that channel is created. These methods formally describe an encoder

$$z = f_{enc}(H), \tag{4.1}$$

and a decoder

$$\hat{H} = f_{dec}(z), \tag{4.2}$$

with z being a latent codeword transmitted along the feedback link. While this method may provide higher rate-distortion performance than fixed codebooks, it considers each instantaneous channel sample to be treated like any other where the entire channel structure needs to be inferred starting with no prior information. In realistic wireless scenarios, this formulation is excessively hard.

The channels that have been taken into consideration in this thesis are not generic random matrices but rather realistic MU-MIMO channels derived under Sionna ray tracing with urban propagation scenarios, as has been explained in Chapters 2 and 3. The channels have a steady form that depends on the environment. As an illustration, similar dominant angular support, delay spread, correlation eigen structure or beam-domain concentration may regularly occur among users in the same area or propagation regime. These long term properties change significantly slower compared to the instantaneous fading realizations. Thus, they are to be regarded as side information, which would lower the uncertainty of the instantaneous channel, in accordance with the information-theoretic and system-design point of view.

Let S denote a long-term statistical descriptor associated with the channel distribution at a given user or scenario. Then the CSI compression problem may instead be formulated as

$$z = f_{enc}(H, S) \tag{4.3}$$

$$\hat{H} = f_{dec}(z, S) \tag{4.4}$$

Under this formulation, the encoder does not need to use its full latent capacity to re-express structural information already contained in S. Instead, it can devote more capacity to representing the innovation or residual part of the channel relative to the long-term structure.

The drive behind such a conditioned representation is very compelling in the current thesis due to three factors. Firstly, the digital twin has made it possible to extract long-term statistics by generating channels repeatedly in the same environment. Secondly, the practical performance goal is MU-MIMO, where maintaining the shape of the channel is considered more significant than reducing the error on a per-element basis. Finally, the scheduler and precoder employed in the paper, which are SUS + PF and RZF respectively, are particularly susceptible to the precision of user channel directions and inter-user relative structure. Thus, enhancing CSI fidelity in an environmental way also has a direct and meaningful implication at the system level.

The statistical descriptors considered in this thesis are the spatial covariance matrix and the power delay profile. The covariance matrix contains dominant spatial eigenspace information and reflects long-term angular and array-domain structure. The PDP captures the coarse delay-domain energy distribution and thereby provides prior information about frequency selectivity and multipath

support. These descriptors are sufficiently informative to be useful, yet sufficiently stable to justify low-rate or infrequent reporting compared with instantaneous CSI.

The explanation will be simple when considering the rate-distortion point of view. When S is given, the conditional entropy of the channel is less than the unconditional entropy. It means that less bits are needed to attain the same distortion level or conversely, the same distortion can be attained with less feedback budget. The intended framework aims to apply this principle to a realistic neural network.

4.2 Baseline: CsiNet Architecture

4.2.1 Angle-Delay Domain Representation of the Channel Matrix

The fundamental autoencoder used in this chapter is built on the CsiNet family of architectures that do not act upon raw frequency-domain MIMO channel matrices, but upon a transformed representation where the channel has more structure and sparsity. The angle-delay domain is very well suited to this task in massive MIMO and OFDM systems, as most of the energy of the channel can be focused into a few bins when a proper transformation is performed..

Suppose that the original wideband channel matrix is denoted by $H \in \mathbb{C}^{N_t \times N_c}$, where N_t denotes the number of transmit antenna ports and N_c denotes the number of subcarriers or effective frequency bins on which feedback is based. The channel can be converted into the angle-delay domain by means of a two-dimensional DFT (or an equivalent space-frequence) transforming it into

$$\widetilde{H} = F_{ang} H F_{del}^{H}, \tag{4.5}$$

with F_{ang} and F_{del} being the transformation matrices along the antenna and frequency axes respectively. In real-life applications, the dominant delay bins are preserved only following the transformation as the remaining bins usually have insignificant energy. When N_c' dominant delay-domain bins are saved, the transformed channel input will become a compact tensor of size $N_t \times N_c'$.

In order to feed this channel into a real-valued neural network, the real and imaginary part are split up and stacked into a two-channel tensor. Hence, the ultimate input to the baseline encoder has the form $2 \times N_t \times N_c'$. Such representation has two benefits. Firstly, it greatly minimizes the input dimensionality because low-energy delay bins are omitted. Secondly, it matches the structure of inputs with the sparse or almost sparse nature of practical wireless channels, which is advantageous in the case of convolutional feature extraction.

In the present thesis, this transformed-domain representation is used both for the CsiNet baseline and for the proposed statistics-conditioned model. This ensures that any observed gain

comes from the use of conditioning information rather than from an arbitrary change of signal representation.

4.2.2 CsiNet Encoder and Decoder

The baseline neural CSI compression model adopted in this thesis follows the CsiNet architecture proposed by Wen, Shih, and Jin in [16]. In this framework, CSI feedback is formulated as an encoder–decoder problem in which the UE compresses the measured channel into a low-dimensional codeword and the gNB reconstructs the channel from that codeword. CsiNet operates on the angle-delay domain representation of the channel, where the channel energy is more concentrated and therefore more suitable for learned compression. For this reason, CsiNet has become one of the most representative reference models in the study of learning-based CSI feedback [16]. The input tensor is first processed through convolutional layers with nonlinear activation so that local spatial-frequency patterns can be detected. These convolutional operations are followed by flattening and a dense bottleneck projection, producing a latent vector $z \in \mathbb{R}^M$, where M is the latent dimension. The compression ratio may be expressed as

$$\gamma = \frac{M}{2N_t N_c'}, \tag{4.6}$$

where the factor of 2 accounts for the real and imaginary channels. Smaller γ corresponds to more aggressive compression.

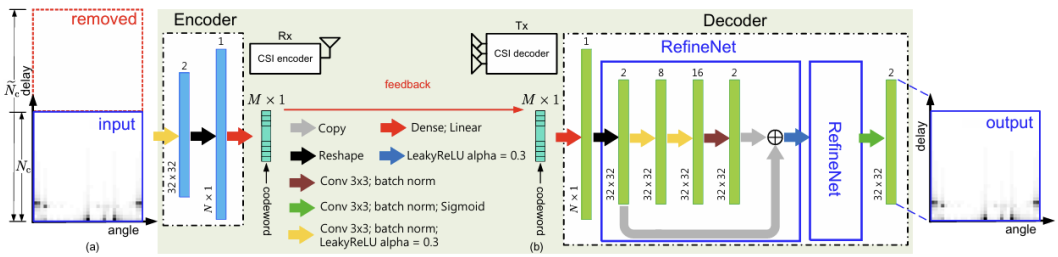

Figure 4.1 CsiNet encoder–decoder architecture adopted from [16]

As shown in Figure 4.1, the input to the network is a cropped angle-delay domain channel representation, where the ineffective delay region is removed and the dominant part of the channel is retained. The encoder initially derives channel features via convolutional processing and subsequently transforms the obtained feature representation into a compact latent vector by using a dense projection. The latent vector matches the compressed CSI code word that will be used in

feedback between the UE and the gNB. On the receiver end, the decoder expands the codeword into a higher-dimensional feature representation, and then restores the channel with a series of convolutional refinement blocks. Originally the refinement blocks were called RefineNet in the CsiNet design and its purpose was to gradually enhance the quality of the reconstructed channel [16].

The importance of this architecture is that it does not use fixed codebook-based CSI representation but a learned non linear compression mechanism. The encoder learns to retain the key channel characteristics at a finite feedback cost rather than restricting the channel to a given beam structure as the decoder learns to recreate the channel based on the compressed codeword. It is thus possible to consider CsiNet as a suitable neural reference model in the current thesis, as it is a popularly accepted blind CSI compression model that can be compared to the suggested approach.

This thesis has the Baseline Encoder based on the concept of CsiNet. Nevertheless, the initial design of CsiNet is based solely on the current channel realization, and it does not explicitly consider long-term environmental structure, including covariance-related information or delay-domain statistics. Therefore, some of the limited latent representation should be used to learn structural information that might have been persistent across channel realizations. The given limitation is also the main source of motivation behind the proposed statistics-conditioned encoder described in the later part of this chapter.

4.3 Proposed Method: Statistics-Conditioned CSI Encoder

4.3.1 Overall Structure

The suggested encoder is based upon a two-stage idea dividing long-term and short-term channel data. The first stage is used to acquire or compress the statistical descriptor of long term, including covariance or PDP. It is slow evolving and thus should be updated rarely. The second stage is used to compress the instantaneous channel realization given the long-term descriptor.

This separation is motivated by the fact that long-term statistics and instantaneous CSI evolve on different timescales. If both are transmitted at the same frequency, the feedback overhead may not be efficient. However, if the long-term statistics are updated only when the environment or user location changes significantly, the average overhead of the statistical side information becomes relatively small compared with the frequent instantaneous CSI updates.

Accordingly, the proposed framework consists of:

1. **Long-term statistics representation**, in which a covariance matrix and/or PDP descriptor is encoded, maintained, or periodically updated.

2. **Conditioned instantaneous CSI compression**, in which the actual per-report CSI autoencoder uses the long-term statistics as conditioning input.

This two-stage interpretation is conceptually important even when Stage 1 statistics are assumed to be already available from the digital twin during offline evaluation. It makes clear that the proposed scheme does not rely on unrealistic always-free side information. Instead, it assumes a realistic asymmetry in update frequency between the environment-dependent statistics and the slot-level instantaneous channel.

The key idea is that the Stage 2 encoder and decoder both receive a conditioning vector or feature map derived from the long-term statistics. Therefore, the latent vector produced by Stage 2 no longer needs to encode the full channel structure blindly. Instead, it can focus on the deviation of the current channel realization from the statistics-implied average structure.

4.3.2 Stage 1: Channel Statistics Autoencoder

The first stage of the proposed framework represents the long-term channel statistics in a compact way. Two main types of statistics are considered: the spatial covariance matrix and the power delay profile.

To perform conditioning on covariance, the covariance matrix is initially estimated based on multiple realizations of the channel in the same scenario or location regime. Because the covariance matrix is Hermitian, there is no need to include all the unique entries but only a compact summary that can be computed using them. It has been observed that direct feeding of the complete covariance matrix to the conditioning path may be too costly in practice, particularly when the dimensions of the antennas increase. One practical solution would then be to reduce the dimensionality of the covariance matrix down to a smaller latent vector z_R with the help of an independent statistics autoencoder or a compact learned embedding network. The statistics embedding represents the major eigen structure and correlation structure necessary to reconstruct CSI.

The average power over delay taps is estimated through repeated realizations in PDP-based conditioning. Because the PDP is one dimensional real valued vector, they are more naturally compact and easy to encode compared to the covariance matrix. To obtain a compressed PDP embedding $\mathbf{z}_p$ a simple fully connected or a 1D convolutional encoder could be used.

The two statistics types also provide information that complements each other. The covariance matrix is particularly useful in relation to spatial structure and antenna-domain correlation, which are significant in beamforming and user separability. Delay-domain energy structure and frequency selectivity is especially well represented by the PDP. In this thesis, both are considered as candidate

or combined conditioning sources, and their usefulness is compared in ablation-style offline evaluations.

A key point in the system interpretation is that the statistics update interval T_{stat} is much larger than the instantaneous CSI reporting interval T_{CSI}. Thus, even if Stage 1 consumes a non-negligible number of bits, its time-averaged overhead can remain modest. This distinction is important when comparing total effective feedback cost with Type 2 codebook feedback.

4.3.3 Stage 2: Conditioned Channel Autoencoder

The second step is key as the main pair of encoder-decoder in the instantaneous CSI. It is an extension of the CsiNet backbone but with added conditioning path. Let c be a trained embedding that was learned using the long-term statistics. Then the conditional encoder can be represented in the form of

$$z = f_{enc}(H, \boldsymbol{c}) \tag{4.7}$$

And the decoder as

$$\widehat{H} = f_{dec}(z, \boldsymbol{c}) \tag{4.8}$$

The conditioning vector c can be introduced in various ways in practice. A simple way would be to use feature concatenation where the conditioning vector is extended and concatenated with intermediate feature maps in the encoder and decoder. Another approach is FiLM-style feature modulation, whereby the conditioning network generates scaling and shifting coefficients that regulate intermediate activations. In this thesis, the main focus is not a single arbitrary injection method, but the environment-conditioned formulation itself. However, when it comes to implementation, feature-wise modulation and concatenation are both obvious options.

When viewed as a signal processor, the conditioned autoencoder must acquire the fact that the same latent vector pattern can be associated with various channel reconstructions based on the long-term statistics. As an illustration, a latent code representing a minor directional error might be decoded in a different way based on a LOS dominant covariance structure than based on a rich NLOS covariance structure. Consequently, the decoder is essentially interpreting the compressed instantaneous information in light of the current environment.

Training may be performed sequentially or jointly. In a sequential approach, the statistics encoder is first trained or fixed so that stable statistics embeddings are available. Then the conditioned Stage 2 autoencoder is trained using those embeddings. In a joint approach, both

statistics representation and instantaneous channel reconstruction can be optimized together. In this thesis, a sequential or semi-sequential strategy is conceptually preferable because it aligns with the timescale separation of the two stages and makes the interpretation of each component clearer.

One of the major goals of the training process in Stage 2 is to reduce reconstruction loss; however, the operational goal is more general: the reconstructed channel must save the geometry pertinent to MU-MIMO. Thus, even other loss terms or measurement criteria associated with directional fidelity, pairwise angular preservation, or latent-space distance preservation can be taken into consideration. It is particularly relevant since, according to the thesis motivation, not every reconstruction error has the same negative impact on MU-MIMO.

4.4 Dataset Construction

The data used in this chapter is based on the RAN digital twin that has been discussed in Chapter 3. It is a significant design decision and it is because it guarantees that the training environment and the environment to be used in future end-to-end system assessment are consistent. The samples of the channel are not created by an unrelated abstract generator, but rather extracted by the same Sionna-based environment-sensitive channel engine that runs the actual system implementation.

For each scenario, UMa-NLOS, UMi-LOS, and UMa-LOS, a set of UE positions and propagation conditions is defined. For each position or local environment condition, large number of channel realizations are generated. These realizations are divided into two conceptual groups. The first group is used to estimate long-term statistics such as covariance and PDP. The second group is used as instantaneous channel samples for training, validation, and testing of the CSI encoder.

The given split is significant since it represents the practical implementation of the suggested approach. Long-term statistics should be derived based on a set of realizations that are representative of the environment whereas the instant channel that needs to be compressed is specific example of that environment. A single realization must not be employed at the same time as the source of statistics and target instantaneous sample so that it could make the conditioning problem trivial.

The dataset in a realistic experimental design should also not leak into the training and test conditions. E.g., when channels occupy the same positions arbitrarily divided among training and test, the model is likely to overfit to that position. So, position-aware or environment-aware dataset partitioning is desirable because it ensures that the model is trained on unobserved realizations or spatial subsets but remains in the same scenario family.

The dataset in this thesis is organized so that each sample consists of:

1. An instantaneous angle-delay-domain channel tensor.
2. A long-term covariance-related descriptor and/or PDP descriptor.
3. A scenario label or scenario grouping.
4. Optional metadata such as SNR or mobility class for analysis.

The provided structure makes it possible to train and test the suggested model in such a manner that it would immediately reflect the target application of statistics-conditioned CSI feedback.

4.5 Offline Performance Evaluation

4.5.1 CsiNet Baseline vs Proposed Method

This section compares the statistics-conditioned encoder (proposed) and the blind-based encoder (CsiNet) using the same compression-ratio configurations. The goal behind this comparison is to eliminate the influence of statistical conditioning on the overall positive impact of learned CSI compression. As the two approaches rely on the same neural encoder-decoder models, the difference in their performances may be mainly explained by the fact that the incorporation of long-term channel statistics into the encoding procedure adds a performance gap to the two methods. The blind CsiNet model uses the traditional CSI compression structure presented in [16], whereas the suggested method builds upon that model by making the encoder conditioned on environment-specific channel descriptors derived using the digital twin.

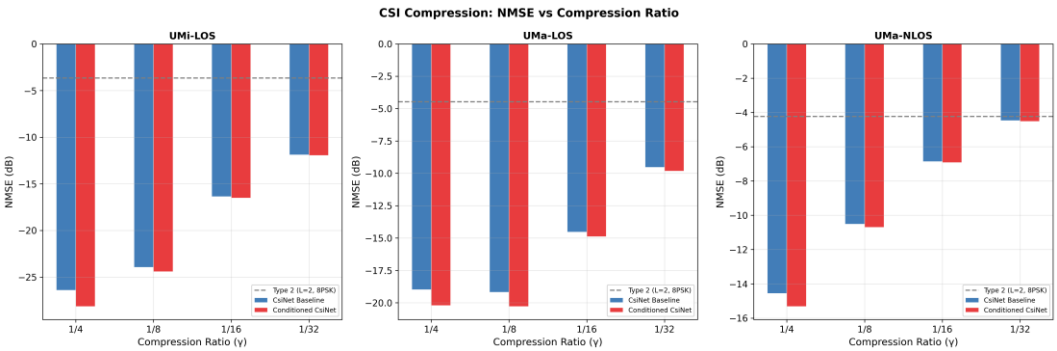

Figure 4.2 NMSE comparison between the CsiNet baseline and the proposed conditioned CsiNet u nder different compression ratios

Figure 4.2 contains the comparison results with the normalized mean square error versus compression ratio for three propagation scenarios, UMi-LOS, UMa-LOS, and UMa-NLOS. All three cases show that the conditional encoder achieves lower NMSE than the blind CsiNet encoder especially at higher compression rates. It means that the proposed approach saves channel information better when it is subjected to the same compression limit. The enhancement can be seen in all the conditions, which implies that the value of long-term statistical conditioning cannot be restricted to one type of propagation.

Under UMi-LOS conditions, both approaches have reasonably high reconstruction results, but the suggested approach always performs better than the blind CsiNet model at any compression ratio. The improvement is particularly evident at compression rates of 1/4 and 1/8 with the conditioned model obtaining significantly less NMSE. This is in accordance with the observation that LOS-dominant propagation has a relatively stable directional structure, which can be efficiently utilized in case long-term statistical side information is present.

A comparable trend can be observed in the UMa-LOS setting. As suggested by the proposed method, the values of NMSE are lower compared to those of the blind CsiNet model at all the selected rates of compression. The difference in performance is still apparent between 1/4 and 1/32, which means that the advantage of statistical conditioning is maintained regardless of whether the environment has more extensive spatial variation compared to the UMi-LOS case. Such an outcome implies that the conditioning approach under consideration can be helpful when applied in highly concentrated micro-cell LOS channels as well as in macro-cell LOS settings where the users are more dispersed.

In the UMa-NLOS scenario, the overall NMSE performance of both methods in this case is inferior compared to the LOS cases as anticipated since the channel structure is more distributed and harder to reconstruct at a limited compression. However, the suggested encoder still offers a steady improvement over the blind CsiNet model. The given outcome is especially significant since it indicates that the suggested solution can be useful even with richer NLOS propagation when the long-term channel structure is not so concentrated but statistically important.

One more helpful finding based on Figure 4.2 is the relationship to the practical Type 2 reference level marked by dashed horizontal line in each of the subplots. At certain operation points especially at moderate compression ratios, the suggested scheme may come close or surpass the blind CsiNet model more obviously than the Type 2 reference. It implies that the obtained gain of the proposed method is not just a numerical decrease in NMSE, but also a practically significant improvement over an operational point of a conventional CSI feedback.

Overall, these results confirm that the proposed encoder improves CSI reconstruction beyond the blind CsiNet framework introduced in [16]. The improvement arises because the proposed method does not rely solely on the instantaneous channel realization. Instead, it exploits long-term

statistical information derived from the same environment-aware RAN twin used throughout this thesis. As a result, the encoder can allocate more of its limited latent representation capacity to the instantaneous variation of the channel, rather than repeatedly encoding structural information that remains relatively stable across realizations. This observation provides the main offline evidence supporting the proposed approach before the end-to-end performance evaluation in Chapter 5.

4.6 Summary

This chapter introduced the proposed statistics-conditioned AI/ML CSI encoder and established the offline evaluation framework. It began by formalizing the limitation of blind CSI compression, namely that instantaneous channels are compressed without exploiting long-term environmental structure. It then presented the CsiNet-style baseline architecture, including the angle-delay-domain channel representation, encoder-decoder structure, and quantized latent feedback path. Next, it introduced the proposed two-stage framework in which slowly varying channel statistics, such as covariance and PDP, are represented in Stage 1 and used as conditioning information for instantaneous CSI compression in Stage 2. The chapter also described how the required training and evaluation dataset is generated from the RAN digital twin, thereby ensuring consistency between the AI model and the final execution environment. Finally, the chapter outlined the offline evaluation strategy, including comparison with the blind CsiNet baseline, comparison with Type 2 codebook feedback, and consistency checks with broader AI/ML CSI feedback trends. It provides the groundwork on which Chapter 5 will be based where the proposed methodology will be evaluated in the overall end-to-end MU-MIMO pipeline of the digital twin.

Chapter 5.

End-to-End Performance Evaluation and Analysis Using the RAN Digital Twin

Chapter gives the end-to-end assessment of the proposed statistics-conditioned CSI encoder implemented in the OAI-based RAN digital twin developed in this thesis. Although the main concern of Chapter 4 has been the performance of the offline reconstruction, the main goal of the current chapter is to discover what gain can be achieved by improving CSI representation in practice. That is important. Practically in 5G NR MU-MIMO system, the CSI is not the final point but a means of an intermediate representation that affects the scheduling, precoding, modulation and coding choice, HARQ behavior, and finally the throughput and fairness perceived by the users. Hence, the value of the suggested encoder should be evaluated not only in terms of normalized mean square error but also in terms of its influence on the whole transmission chain.

The chapter follows the evaluation logic that is progressive. Firstly, the experimental environment and the methodology are presented, such as the common radio parameters, comparison baselines, and the scenario matrix. Secondly the three traditional operating modes i.e. Type 1 SU-MIMO, Type 2 SU-MIMO, and Type 2 MU-MIMO are compared to determine the practical constraints of the available CSI representations and define the performance region where enhanced CSI is likely to have the greatest impact. Third, the suggested statistics-conditioned encoder is introduced into the MU-MIMO feedback loop and its end-to-end gains over the traditional Type 2 MU-MIMO baseline are examined. Lastly, the findings are explained more broadly, such as the difference between genies and assisted operation, how statistical conditioning is helpful, and the nonlinear relationship between the offline reconstruction measurements and ultimate throughput.

5.1 Experimental Environment and Methodology

5.1.1 Experimental Setup

All experiments in this chapter are conducted in an FR1 Band 78 downlink configuration centered at 3.5 GHz with subcarrier spacing of 30 kHz and a bandwidth of 106 PRBs, corresponding to approximately 40 MHz. The gNB is configured with a 4T4R dual-polarized antenna structure, corresponding to $N_x = 2$, $N_y = 1$, and cross-polarized operation with $XP = 2$, yielding four effective antenna ports. This 4-port configuration is compatible with the real-world Type II-focused

CSI acquisition and MU-MIMO deployment presented in Chapter 3. The CSI-RS is set to operate at a periodicity of 20 slots, and the time spent on every experiment is enough (on the scale of thousands of slots) to get stable averages of throughput, BLER and scheduling behavior.

The protocol stack is implemented on the basis of the OAI-based RAN digital twin whereas the propagation channels are created by means of the environment-sensitive channel engine based on the Sionna as described before. The cases discussed include the UMa-NLOS, UMi-LOS, and the UMa-LOS, and the same radio numerology and feedback structure are used in all the methods, such that the comparison separates the impact of CSI representation and scheduling mode. The default MU-MIMO scheduler is semiprime user selection with proportional-fair weighting and the downlink precoder is zero-forced regularization. Thus, the observed variations in performances in MU-MIMO mode can be directly explained by how the accuracy of CSI influences the pairing of users, effective channel inversion, and interference suppression.

The proposed method is inserted into the system as a replacement or enhancement of the practical CSI representation used by the MU-MIMO scheduler and precoder. Long-term statistical descriptors are obtained from the same digital twin environment, and the conditioned encoder is used to produce the compressed channel representation used for practical scheduling and precoding decisions. In this sense, the evaluation is fully end-to-end: the proposed method is not judged only by offline reconstruction but by the behavior of the complete OAI-Sionna execution pipeline.

5.1.2 Comparison Targets

Five principal operating modes are considered in the end-to-end comparison.

The first of these is Type 1 SU-MIMO, where the codebook-based CSI feedback is used along with single-user MIMO transmission. It is the most conservative practical operating point of the compared methods, as it uses a relatively compact CSI representation and does not utilize multi-user spatial multiplexing.

It is the second, Type 2 SU-MIMO, where the scheme employs Type 2 codebook-based CSI feedback in spite of the fact that it is used during a single-user transmission mode. The present configuration can isolate the benefits of enhancing the quality of the CSI representation in SU-MIMO without adding the complexity of multi-user interference.

The third is Type 2 MU-MIMO, which will employ Type 2 codebook-based CSI feedback, as well as the practical MU-MIMO transmission chain that is applied to the proposed RAN twin. In this mode, the grouping of users is done by the implemented scheduling algorithm and the downlink transmission is done by the implemented precoding algorithm. The configuration is the most powerful traditional CSI-feedback-based multi-user operation point discussed in this thesis.

The fourth is the Baseline Encoder, which replaces the conventional codebook-based CSI representation in the MU-MIMO transmission chain with a blind AI/ML-based CSI encoder. This encoder uses only the instantaneous channel realization as input and does not explicitly exploit long-term channel statistics. Its role in the comparison is to isolate the benefit of learned CSI compression itself, separate from the additional benefit of statistical conditioning.

The fifth is the Proposed Method, in which the practical MU-MIMO transmission chain uses the proposed statistics-conditioned AI/ML CSI encoder. In this case, long-term channel descriptors extracted from the RAN digital twin, such as covariance-related or delay-domain statistical information, are used as side information for CSI compression. This configuration is the main contribution of the thesis and is evaluated against both the conventional codebook-based operating points and the blind AI/ML encoder.

The comparison structure is significant since it helps to identify the role played by each design option to be easily identifiable. The comparison of Type 1 SU-MIMO and Type 2 SU-MIMO shows the implication of enhancing traditional CSI accuracy when used in a single-user environment. The comparison of Type 2 SU-MIMO and Type 2 MU-MIMO indicates the benefit and constraint of multi-user communication at feasible Type 2 feedback. The comparison of Type 2 MU-MIMO and Baseline Encoder demonstrates how using the blind neural compression of CSI feedback can affect structured codebook-based feedback. Lastly, the comparison between the Baseline Encoder and the proposed method indicates the extra gain achieved through the incorporation of long-term channel statistics in the CSI encoder.

5.1.3 Performance Metrics

The evaluation of the performances of the compared transmission schemes is based on three system level metrics: BLER, average MCS index, and cell downlink throughput.

The block error rate (BLER) is the first metric, which represents the reliability of the transmission link. Because BLER also shows how frequently transmitted transport blocks cannot be properly decoded, it is a strong indicator of whether or not a particular representation of the CSI and type of transmission can sustain constant communication at the given propagation conditions.

The next measurement is the average MCS index, which can be used to determine the perceived quality of effective transmission of the system. On average, a higher MCS will usually mean that the existing CSI as well as the subsequent scheduling and precoding choices enable the transmitter to choose more aggressive levels of modulation and coding. This metric is thus helpful in explaining how the quality of CSI is converted into actual link adaptation behavior.

The third metric is the cell downlink throughput, which measures the aggregate throughput achieved at the cell level. This is the most important performance indicator for the present study because the main objective of MU-MIMO is to improve overall cell capacity through simultaneous multi-user transmission. Accordingly, cell throughput provides the most direct measure of the practical gain obtained from improved CSI representation and multi-user transmission capability.

Combined, these three metrics offer a brief albeit adequate picture of the performance of the system. BLER refers to transmission reliability, average MCS is linked to proper link adaptation behaviour, and cell downlink throughput represents the ultimate system-level advantage that is obtained in comparison with the considered transmission schemes.

5.2 End-to-End Performance Under Different Propagation Scenarios

This part shows the overall performance of the five transmission schemes that have been compared in three typical propagation environments, namely UMi-LOS, UMa-NLOS, and UMa-LOS. In each case, the same system-level measures and execution framework are used such that the effect of the CSI representation and transmission mode may be tested under similar experimental settings. The five schemes being explored include Type 1 SU-MIMO, Type 2 SU-MIMO, Type 2 MU-MIMO, Baseline Encoder, which involves blind CsiNet-style CSI compression, and Proposed Encoder, which involves statistics-conditioned CSI compression.

5.2.1 End-to-End Performance in the UMi-LOS Scenario

The end-to-end performance of the UMi-LOS situation is illustrated in Figure 5.1. Because the traditional performance ordering of Type 1 SU-MIMO, Type 2 SU-MIMO and Type 2 MU-MIMO have already been established in Section 3.5, this section will focus on the extra gain of using the blind CsiNet-based MU-MIMO scheme and the proposed conditioned CsiNet-based MU-MIMO scheme.

Both AI-based MU-MIMO schemes are better than the conventional Type 2 MU-MIMO operating point in the BLER results of the low-UE region, but they can be compared in the high-UE region. In most numbers of UEs, the proposed conditioned encoder tends to have the lowest BLER or be tied with the highest performing scheme. There is a similar pattern in the average MCS index where the two AI-based schemes are found to be higher or close to the conventional Type 2 MU-MIMO mode, with the proposed encoder usually providing the most favorable values.

The most obvious benefit is on the throughput of cell downlink. The two learned CSI feedback mechanisms are better than traditional Type 2 MU-MIMO mode, and the suggested conditioned

encoder always yields the best throughput among the five systems compared. It can be seen and it is moderate, that the gain over the blind CsiNet scheme means that in the UMi-LOS case, the prevailing channel structure may be sufficiently well approximated using only the instantaneous channel realization. However, the proposed encoder also performs the best overall, suggesting that statistical conditioning is still advantageous even with a highly directional LOS condition.

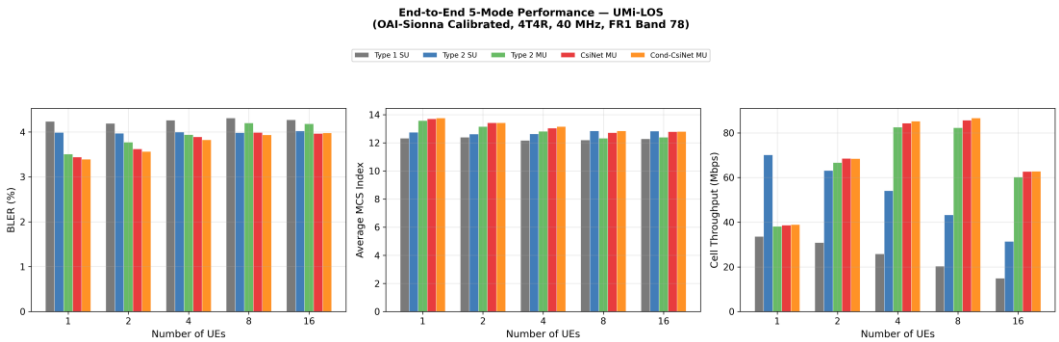

Figure 5.1 End-to-end 5-mode performance comparison in the UMi-LOS scenario

5.2.2 End-to-End Performance in the UMa-NLOS Scenario

Figure 5.2 shows the end-to-end performance in the UMa-NLOS situation. In this case, the benefit of the AI-based CSI feedback schemes is even clearer than that in UMi-LOS.

Both learned MU-MIMO schemes are significantly better than the conventional Type 2 MU-MIMO mode in terms of BLER, and the suggested conditioned encoder has the least error rate under all UE settings that have been tested. The same ranking can be seen in both average MCS index and it shows that the proposed encoder always produces the largest values, which is followed by the blind CsiNet scheme, whereas the traditional Type 2 MU-MIMO mode is lower than each of the two learned schemes.

The cell throughput results show the strongest improvement. The blind CsiNet-based MU-MIMO scheme already improves substantially over conventional Type 2 MU-MIMO, and the proposed conditioned encoder gives a further consistent gain on top of that. The advantage of the proposed method is especially visible in the low- and medium-UE region, where it achieves the highest cell throughput among all compared schemes. This indicates that in the UMa-NLOS scenario, long-term statistical conditioning provides a particularly meaningful improvement over blind learned compression.

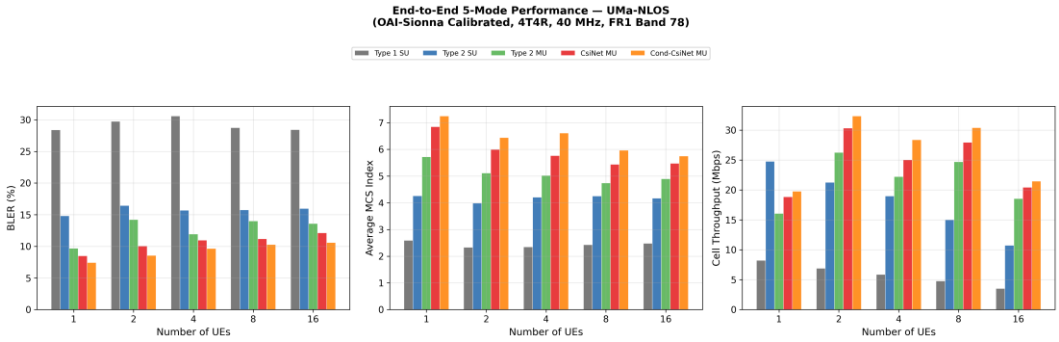

Figure 5.2 End-to-end 5-mode performance comparison in the UMa-NLOS scenario

5.2.3 End-to-End Performance in the UMa-LOS Scenario

The end-to-end performance in the UMa-LOS scenario is shown in Fig. 5.Z. As in the previous two scenarios, the learned CSI feedback schemes provide additional gain beyond the conventional Type 2 MU-MIMO mode.

In BLER, the blind and conditioned CsiNet schemes both improve upon the conventional MU-MIMO mode, with the proposed encoder providing the best or near-best performance over the tested UE range. The same tendency appears in the average MCS index, where the proposed encoder consistently remains at the top or statistically tied with the best scheme.

Considering the cell throughput, each of these two AI-based schemes is superior to the traditional Type 2 MU-MIMO scheme, and the proposed conditioned encoder performs the best of the five schemes. It can be seen that the gain over blind CsiNet is present throughout the whole UE range, though it is not as high as in the UMa-NLOS case. It means that the suggested encoder is also useful in the UMa-LOS situation, but the extra benefit provided by statistical conditioning is less than in the more populated NLOS situation.

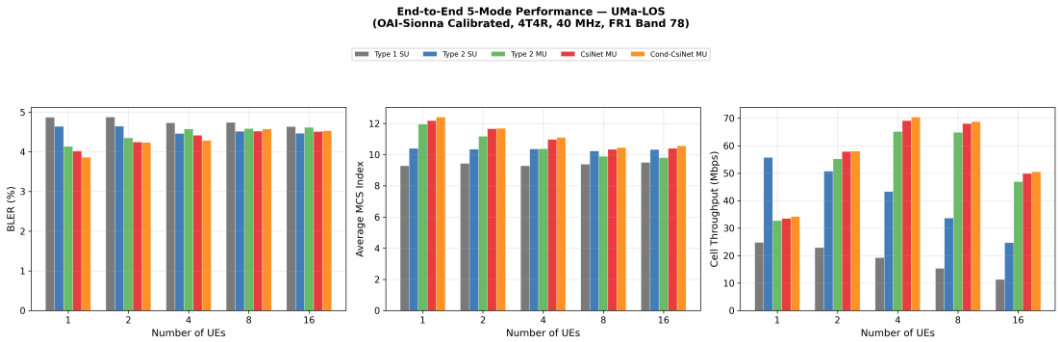

Figure 5.3 End-to-end 5-mode performance comparison in the UMa-LOS scenario

5.3 Cross-Scenario Performance Analysis

In all three situations, the suggested conditioned encoder is the most optimal in the overall performance of the five discussed schemes. As the typical sequence between Type 1 SU-MIMO, Type 2 SU-MIMO, and Type 2 MU-MIMO has already been introduced in Section 3.5, the most significant conclusion in this context is that when conventional Type 2 MU-MIMO is substituted first with blind learned CSI compression and then with the proposed statistics-conditioned learned CSI compression, an additional gain is achieved.

As shown in the results, the proposed encoder always performs better than the blind CsiNet-based MU-MIMO scheme in BLER, average MCS index, and throughput at the cell downlink. Nevertheless, the extent of such improvement varies depending on the situation. The gain of the three environments is the highest in the UMa-NLOS scenario. On UMi-LOS and UMa-LOS, the proposed encoder remains the top performer, though the difference with blind CsiNet is less pronounced.

One possible explanation of this discrepancy is the underlining channel structure. It is observed that in LOS-dominant conditions and particularly UMi-LOS, the channel is more directional and focused, and hence, a blind encoder is capable of capturing a significant portion of the useful structure by simply using the instantaneous channel. On an UMa-LOS, the same trend persists but the geometry of the macro-cells leads to slightly greater variability. Conversely, in UMa-NLOS the channel has more dispersed characteristics and more rich multipath and wider angular support. In this situation, blind instantaneous-channel compression is inefficient since the encoder needs to learn more of the environment-dependent structure in every sample. As the suggested method alleviates this load, it provides long-term channel statistics as side information, enabling the encoder to concentrate more of its restricted latent capacity to the instantaneous variation that is most important.

Another important point is that the gain of the proposed encoder is reflected not only in offline reconstruction metrics, but also in the practical end-to-end transmission chain. The proposed method improves BLER and average MCS, which indicates more reliable link adaptation and more accurate transmitter-side channel representation. At the same time, it improves cell throughput, which means that the benefit propagates into actual scheduling and precoding behavior. Therefore, the advantage of the proposed encoder should be understood as a system-level improvement rather than as a purely reconstruction-level gain.

On the whole, the cross-scenario findings confirm the main claim of the thesis. The suggested statistics-conditioned encoder is always better than the traditional codebook-based CSI feedback and even the blind learned CSI compression, and it performs best in the UMa-NLOS case with the most distributed channel structure and largest demand on long-term statistical side information.

5.4 Summary

The results of this chapter highlight three important points regarding practical CSI feedback for MU-MIMO transmission.

First, the comparison between the Baseline Encoder and the Proposed Encoder shows that the key gain of the proposed method does not come merely from replacing structured codebook feedback with a neural encoder. Both of these methods use learned CSI compression. The essential difference is that the Proposed Encoder incorporates long-term channel statistics as side information, whereas the Baseline Encoder relies only on the instantaneous channel realization. The observed performance gap therefore supports the main argument of the thesis: the CSI compression problem becomes more effective when it is formulated as a statistics-conditioned channel-representation task rather than as a blind instantaneous-channel compression task.

Further, as evidenced by the findings, offline reconstruction quality and end-to-end throughput are correlated but not strictly so. A slight gain in reconstruction of CSI can also yield significant throughput benefits when it alters the choice of the scheduler on how they group users of the MU-MIMO or enhances the conditioning of the effective multi-user channel matrix that is employed in precoding. On the other hand, there might be some reconstruction enhancements that could have no significant throughput impact in regions where the system is limited by some other constraints. The above provides the reason behind the need to validate end-to-end. Offline measures still help in explaining the working of encoders but cannot on their own reflect the entire system utility of better CSI representation in an operational MU-MIMO chain.

Thirdly, despite the consistent findings of the proposed method, there are still various limitations of the current study. The deployment has been based on a 4T4R-scale system as opposed to bigger massive-MIMO systems. Its execution environment is in non-real-time which is fine with

controlled experiments, but it is not yet available to emulate them in a deployment style. Moreover, the learned CSI encoders discussed in this paper focus on a CSI Net-like construction instead of more developed classes of models. However, it is evident that the statistics of channels over long periods could enhance the practical representation of CSI over both the common codebook-based feedback and blind neural compression. It gives a solid foundation to future research on larger sizes of antennas, more efficient encoder designs, and explicit runtime schemes to ensure long-term statistics preservation and upgrading.

Chapter 6.

Conclusion

The present thesis explored operational implementation of CSI representation in 5G NR MIMO transmission considering real-world wireless conditions, especially focusing on the bottleneck of the CSI in MU-MIMO operation. The research was initiated by the fact that even though MU-MIMO has a great potential in achieving significant cell-throughput improvement relative to SU-MIMO, it has very limited performance in practice due to the availability of the CSI at the transmitter. In a limited-feedback environment, inaccuracies in CSI have implications on both beamforming performance, as well as user grouping, interference cancellation, link adaptation and finally end-to-end throughput. Such a situation makes CSI representation an important issue in practical MU-MIMO system design.

In order to solve this issue, the thesis initially defined a realistic experimental environment that was based on a digital twin of an OAI-based RAN. The created platform combines the OAI protocol stack, a Sionna-based environment-aware channel engine, and a centralized control structure to manage scenarios and repeatable experimentation. This framework allowed practical CSI acquisition in a 4x4 MIMO setup, and extended the downlink transmission chain to accommodate the processing of CSI in Type 2, MU-MIMO scheduling, and regularized zero-forcing precoding. Consequently, the platform could assess the MIMO performance dependent on CSI in realistic propagation conditions in addition to acting as the source of channel statistics over the long term.

Another significant contribution of the thesis was to develop a feasible transmission chain that could be used to evaluate comparatives. Rather than restricting the analysis to the reconstruction of offline channels, the paper developed an end to end system that is realistic where the CSI quality has a direct influence on scheduling and precoding behavior. Specifically, the thesis used semi-orthogonal user selection along with proportional-fair scheduling and regularized zero-forcing precoding. The design allowed seeing the effect of the representation of CSI on the performance of MU-MIMO via two system-level mechanisms, i.e., multi-user grouping reliability and interference cancellation efficiency following the precoding process.

In addition to this practical system, the thesis offered a statistics-conditioned AI/ML CSI encoder. Besides the traditional codebook-based feedback, and in contrast to the blind neural CSI compression, the suggested approach explicitly uses long-term channel descriptors (covariance-related and delay-domain statistical information) in the CSI coding process. The key concept behind it was that realistic wireless channels do not correspond to any isolated random matrix but to environment-conditioned realization with persistent structure. In case that long-term structure has

been given to the encoder, then the limited representation capacity of the encoder can be used more effectively by emphasizing on the instantaneous change as opposed to the long-term channel model.

According to the offline assessments, the suggested approach is more efficient in representing CSI as compared to the conventional structured feedback and blind learned compression given similar feedback limitations. Even more significantly, the end-to-end measurements indicated that this enhancement still makes sense after it has been through the entire transmission chain. In the considered configurations, i.e., UMi-LOS, UMa-NLOS, and UMa-LOS, the proposed scheme was always the most effective one of all the schemes that were compared. The gains that were observed were not only reflected in better quality of channel representation, but also in better quality of MU-MIMO user grouping, better quality of practical precoding, and higher sum cell throughput.

The scenario-based analysis also showed that the role of CSI representation is scenario-dependent. In LOS-dominated cases, the greatest problem is maintaining fine directional differences between users who could otherwise have very similar channel characteristics. In more complex NLOS situations, the biggest problem is maintaining distributed support of channels and more general multi-user geometry. It was demonstrated that the suggested approach would be useful in both situations but the principle of gain varied depending on the channel structure. This indicates that the utility of statistical conditioning does not depend on a particular type of scenario, but spans several realistic scenarios of urban propagation.

One further significant implication of the thesis is that the quality of offline reconstruction and end-to-end throughput are correlated but not related in a linear manner. A slight increase in CSI reconstruction could give rise to a significant throughput enhancement if it alters the choice of user grouping of the scheduler or enhances the quality of the effective multi-user channel matrix applied by the precoder. Conversely, certain reconstruction improvements might result in small throughput variations in operating fields where other limits dominate. The result shows that realistic end-to-end measurements are required in experimental research on CSI feedback. Reconstruction metrics cannot be the only measure of evaluating a CSI encoder if the final objective is MU-MIMO system performance.

Overall, the thesis supports the following main conclusion: practical CSI feedback for MU-MIMO can be improved significantly when the feedback problem is formulated not as blind instantaneous-channel compression, but as environment-conditioned channel representation. By using long-term channel statistics extracted from a realistic RAN digital twin, the proposed method improves the practical usefulness of CSI under limited feedback and thereby enhances the end-to-end performance of MU-MIMO transmission.

Even with such contributions, various limitations are present. The existing implementation does not involve a larger massive-MIMO array but rather a 4T4R-scale setting. The execution environment is non-real-time, which is acceptable in controlled experiments, however it cannot be

used to emulate in deployments yet. Besides, the learned encoder discussed in this thesis is based on CsiNet-style architecture, and additional more sophisticated families of models can offer further enhancement. The way the long-term statistical side information is handled is also idealized as the maintenance and update strategy of the statistics has not been totally optimized in the runtime system.

These constraints indicate a number of areas of future research. Initially, the given model must be expanded to more large-scale antennas and greater bandwidth systems to assess whether it is valuable in more challenging massive-MIMO scenarios. Secondly, more efficient neural architectures such as transformer-based or hybrid geometry-aware architectures can be employed to enhance the efficiency of CSI representation. Thirdly, specific time strategies to acquire and update long term statistics ought to be included in the digital twin to model the overhead of statistical side information more accurately. Lastly, the suggested framework could also be applied outside the existing downlink scenario towards more general environment-aware transmission design issues in future 5G-Advanced and 6G systems.

To sum up, it has been established in this thesis that realistic environmental awareness of CSI representation is an effective methodology of enhancing practical MU-MIMO operation. The work presented a full methodology of how to link channel representation studies to end-to-end wireless system performance by integrating an OAI-based RAN digital twin with a practical scheduling and precoding chain and a statistics-conditioned AI/ML CSI encoder. The findings show that long term channel knowledge will be beneficial to the feedback procedure when integrated properly into the process in question, as opposed to the conventional codebook-based CSI feedback or the blind neural compression. It is this potential that makes environment-conditioned CSI feedback a good avenue to follow in future practical MIMO systems.

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